

Answers (to Selected Problems)

1. (i) $1 = a^{-1}a = a^{-1}(ax) = (a^{-1}a)x = 1 \cdot x = x$.
 (iii) If $x^2 = y^2$, then $0 = x^2 - y^2 = (x - y)(x + y)$, so either $x - y = 0$ or $x + y = 0$, that is, either $x = -y$ or $x = y$.
 (vi) Replace y by $-y$ in (iv).
2. One step requires dividing by $x - y = 0$.
3. (i) $a/b = ab^{-1} = (ac)(b^{-1}c^{-1}) = (ac)(bc)^{-1}$ (by (iii)) $= ac/bc$.
 (ii) $(ad + bc)/(bd) = (ad + bc)(bd)^{-1} = (ad + bc)(b^{-1}d^{-1})$ (by (iii)) $= ab^{-1} + cd^{-1} = a/b + c/d$.
 (iii) $ab(a^{-1}b^{-1}) = (a \cdot a^{-1})(b \cdot b^{-1}) = 1$, so $a^{-1} \cdot b^{-1} = (ab)^{-1}$.
 (v) $(a/b)/(c/d) = (a/b)(c/d)^{-1} = (a \cdot b^{-1})(c \cdot d^{-1})^{-1} = (a \cdot b^{-1})(c^{-1} \cdot d) = ad(b^{-1} \cdot c^{-1}) = ad(bc)^{-1} = (ad)/(bc)$.
4. (i) $x < -1$.
 (iii) $x > \sqrt{7}$ or $x < -\sqrt{7}$.
 (v) All x , since $x^2 - 2x + 2 = (x - 1)^2 + 1$.
 (vii) $x > 3$ or $x < -2$, since 3 and -2 are the roots of $x^2 - x - 6 = 0$.
 (ix) $x > \pi$ or $-5 < x < 3$.
 (xi) $x < 3$.
 (xiii) $0 < x < 1$.
5. (i) $b - a$ and $d - c$ are in P , so $(b - a) + (d - c) = (b + d) - (a + c)$ is in P . Thus, $b + d > a + c$.
 (iii) Using (ii), $-c < -d$; then (i) implies that $a + (-c) < b + (-d)$.
 (v) $(b - a)$ and $-c$ are in P , so $-c(b - a) = ac - bc$ is in P , that is, $ac > bc$.
 (vii) Using (iv), $a > 0$ and $a < 1$, so $a^2 < a$.
 (ix) Substitute a for c and b for d in (viii).
9. (i) $\sqrt{2} + \sqrt{3} - \sqrt{5} + \sqrt{7}$.
 (iii) $|a + b| + |c| - |a + b + c|$.
 (v) $\sqrt{2} + \sqrt{3} + \sqrt{5} - \sqrt{7}$.
10. (i) a if $a \geq -b$ and $b \geq 0$;
 $-a$ if $a \leq -b$ and $b \leq 0$;
 $a + 2b$ if $a \geq -b$ and $b \leq 0$;
 $-a - 2b$ if $a \leq -b$ and $b \geq 0$.
 (iii) $x - x^2$ if $x \geq 0$;
 $-x - x^2$ if $x \leq 0$.
11. (i) $x = 11, -5$.
 (iii) $-6 < x < -2$.
 (v) No x (the distance from x to 1 plus the distance from x to -1 is at least 2).
 (vii) $x = 1, -1$.
12. (i) $(|xy|)^2 = (xy)^2 = x^2y^2 = |x|^2|y|^2 = (|x| \cdot |y|)^2$; since $|xy|$ and $|x| \cdot |y|$ are both ≥ 0 , this proves that $|xy| = |x| \cdot |y|$.
 (iii) $|x|/|y| = |x| \cdot |y|^{-1} = |x| \cdot |y^{-1}|$ (by (ii)) $= |xy^{-1}|$ (by (i)) $= |x/y|$.
 (v) It follows from (iv) that $|x| = |y - (y - x)| \leq |y| + |y - x|$, so $|x| - |y| \leq |x - y|$.
 (vii) $|x + y + z| \leq |x + y| + |z| \leq |x| + |y| + |z|$. If equality holds, then $|x + y| = |x| + |y|$, so x and y have the same sign. Moreover, z must

have the same sign as $x + y$, so x , y , and z must all have the same sign (unless one is 0).

CHAPTER 2

1. (i) Since $1^2 = 1 \cdot (2) \cdot (2 \cdot 1 + 1)/6$, the formula is true for $n = 1$. Suppose that the formula is true for k . Then

$$\begin{aligned} 1^2 + \cdots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= \frac{(k+1)}{6} [k(2k+1) + 6(k+1)] \\ &= \frac{(k+1)}{6} [(k+2)(2k+3)] \\ &= \frac{(k+1)(k+2)(2[k+1]+1)}{6}, \end{aligned}$$

so the formula is true for $k+1$.

2. (i)

$$\begin{aligned} \sum_{i=1}^n (2i-1) &= 1 + 3 + 5 + \cdots + (2n-1) \\ &= 1 + 2 + 3 + \cdots + 2n - 2(1 + \cdots + n) \\ &= \frac{(2n)(2n+1)}{2} - n(n+1) \\ &= n^2. \end{aligned}$$

5. (a) Since

$$1 + r = \frac{1 - r^2}{1 - r},$$

the formula is true for $n = 1$. Suppose that

$$1 + r + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

Then

$$\begin{aligned} 1 + r + \cdots + r^n + r^{n+1} &= \frac{1 - r^{n+1}}{1 - r} + r^{n+1} \\ &= \frac{1 - r^{n+1} + r^{n+1}(1 - r)}{1 - r} \\ &= \frac{1 - r^{n+2}}{1 - r}. \end{aligned}$$

- (b)

$$\begin{aligned} S &= 1 + r + \cdots + r^n \\ rS &= r + \cdots + r^n + r^{n+1}. \end{aligned}$$

Thus

$$S(1 - r) = S - rS = 1 - r^{n+1},$$

so

$$S = \frac{1 - r^{n+1}}{1 - r}.$$

6. (i) From

$$(k+1)^4 - k^4 = 4k^3 + 6k^2 + 4k + 1, \quad k = 1, \dots, n$$

we obtain

$$(n+1)^4 - 1 = 4 \sum_{k=1}^n k^3 + 6 \sum_{k=1}^n k^2 + 4 \sum_{k=1}^n k + n,$$

so

$$\begin{aligned} \sum_{k=1}^n k^3 &= \frac{(n+1)^4 - 1 - 6 \frac{n(n+1)(2n+1)}{6} - 4 \frac{n(n+1)}{2} - n}{4} \\ &= \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4}. \end{aligned}$$

- (iii) From

$$\frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)}, \quad k = 1, \dots, n$$

we obtain

$$1 - \frac{1}{n+1} = \sum_{k=1}^n \frac{1}{k(k+1)}.$$

8. 1 is either even or odd, in fact it is odd. Suppose n is either even or odd; then n can be written either as $2k$ or $2k+1$. In the first case $n+1 = 2k+1$ is odd; in the second case $n+1 = 2k+1+1 = 2(k+1)$ is even. In either case, $n+1$ is either even or odd. (Admittedly, this looks fishy, but it is really correct.)
9. Let B be the set of all natural numbers l such that $n_0 - 1 + l$ is in A . Then 1 is in B , and $l+1$ is in B if l is in B , so B contains all natural numbers, which means that A contains all natural numbers $\geq n_0$.
12. (a) Yes, for if $a+b$ were rational, then $b = (a+b) - a$ would be rational. If a and b are irrational, then $a+b$ could be rational, for b could be $r - a$ for some rational number r .
- (b) If $a = 0$, then ab is rational. But if $a \neq 0$, then ab could not be rational, for then $b = (ab) \cdot a^{-1}$ would be rational.
- (c) Yes; for example, $\sqrt[3]{2}$.
- (d) Yes; for example, $\sqrt{2}$ and $-\sqrt{2}$.
13. (a) Since

$$\begin{aligned} (3n+1)^2 &= 9n^2 + 6n + 1 = 3(3n^2 + 2n) + 1, \\ (3n+2)^2 &= 9n^2 + 12n + 4 = 3(3n^2 + 4n + 1) + 1, \end{aligned}$$

it follows that if k^2 is divisible by 3, then k must also be divisible by 3. Now suppose that $\sqrt{3}$ were rational, and let $\sqrt{3} = p/q$ where p and

q have no common factor. Then $p^2 = 3q^2$, so p^2 is divisible by 3, so p must be. Thus, $p = 3p'$ for some natural number p' , and consequently $(3p')^2 = 3q^2$, or $3(p')^2 = q^2$. Thus, q is also divisible by 3, a contradiction.

The same proofs work for $\sqrt{5}$ and $\sqrt{6}$, because the equations

$$\begin{aligned}(5n+1)^2 &= 25n^2 + 10n + 1 = 5(5n^2 + 2n) + 1, \\(5n+2)^2 &= 25n^2 + 20n + 4 = 5(5n^2 + 4n) + 4, \\(5n+3)^2 &= 25n^2 + 30n + 9 = 5(5n^2 + 6n + 1) + 4, \\(5n+4)^2 &= 25n^2 + 40n + 16 = 5(5n^2 + 8n + 3) + 1,\end{aligned}$$

and the corresponding equations for numbers of the form $6n+m$, show that if k^2 is divisible by 5 or 6, then k must be. The proof fails for $\sqrt{4}$, because $(4n+2)^2$ is divisible by 4. (For precisely this reason this proof cannot be used to show that in general $\sqrt{a}$ is irrational if a is not a perfect square—we have no guarantee that $(an+m)^2$ might not be a multiple of a for some $m < a$. Actually, this assertion is true, but the proof requires the information in Problem 17.)

(b) Since

$$(2n+1)^3 = 8n^3 + 12n^2 + 6n + 1 = 2(4n^3 + 6n^2 + 3n) + 1,$$

it follows that if k^3 is even, then k is even. If $\sqrt[3]{2} = p/q$ where p and q have no common factors, then $p^3 = 2q^3$, so p^3 is divisible by 2, so p must be. Thus, $p = 2p'$ for some natural number p' , and consequently $(2p')^3 = 2q^3$, or $4(p')^3 = q^3$. Thus, q is also even, a contradiction.

The proof for $\sqrt[3]{3}$ is similar, using the equations

$$\begin{aligned}(3n+1)^3 &= 27n^3 + 27n^2 + 9n + 1 = 3(9n^3 + 9n^2 + 3n) + 1, \\(3n+2)^3 &= 27n^3 + 54n^2 + 36n + 8 = 3(9n^3 + 18n^2 + 12n + 2) + 2.\end{aligned}$$

19. If $n = 1$, then $(1+h)^n = 1+nh$. Suppose that $(1+h)^n \geq 1+nh$. Then

$$\begin{aligned}(1+h)^{n+1} &= (1+h)(1+h)^n \geq (1+h)(1+nh), \quad \text{since } 1+h > 0 \\&= 1 + (n+1)h + nh^2 \geq 1 + (n+1)h.\end{aligned}$$

For $h > 0$, the inequality follows directly from the binomial theorem, since all the other terms appearing in the expansion of $(1+h)^n$ are positive.

CHAPTER 3

1. (i) $(x+1)/(x+2)$; the expression $f(f(x))$ makes sense only when $x \neq -1$ and $x \neq -2$.
 (iii) $1/(1+cx)$ (for $x \neq -1/c$ if $c \neq 0$).
 (v) $(x+y+2)/(x+1)(y+1)$ (for $x, y \neq -1$).
 (vii) Only $c = 1$, since $f(x) = f(cx)$ implies that $x = cx$, and this must be true for at least one $x \neq 0$.
2. (i) $y \geq 0$ and rational, or $y \geq 1$.
 (iii) 0.
 (v) $-1, 0, 1$.

3. (i) $\{x : -1 \leq x \leq 1\}$.
 (iii) $\{x : x \neq 1 \text{ and } x \neq 2\}$.
 (v) $\emptyset$.
4. (i) 2^{2y} .
 (iii) $2^{2 \sin t} + \sin(2t)$.
5. (i) $P \circ S$.
 (iii) $S \circ S$.
 (v) $P \circ P$.
 (vii) $S \circ S \circ S \circ P \circ P \circ P \circ S$.
11. (a) y .
 (b) $H(y)$.
 (c) $H(y)$.
12. (a)

	even	odd
even	even	neither
odd	neither	odd

(b)

	even	odd
even	even	odd
odd	odd	even

(c)

	f even	f odd
g even	even	even
g odd	even	odd

- (d) Let $g(x) = f(x)$ for $x \geq 0$ and define g arbitrarily for $x < 0$.
21. (a) Let $g(x) = h(x) = 1$ and let f be a function for which $f(2) \neq f(1) + f(1)$. Then $f \circ (g + h) \neq f \circ g + f \circ h$.
 (b) $[(g + h) \circ f](x) = (g + h)(f(x)) = g(f(x)) + h(f(x)) = (g \circ f)(x) + (h \circ f)(x) = [(g \circ f) + (h \circ f)](x)$.
 (c) $\frac{1}{f \circ g}(x) = \frac{1}{f(g(x))} = \frac{1}{f}(g(x)) = \left(\frac{1}{f} \circ g\right)(x)$.
 (d) Let $g(x) = 2$ and let f be a function for which $f(\frac{1}{2}) \neq 1/f(2)$. Then $1/(f \circ g) \neq f \circ (1/g)$.

CHAPTER 4

1. (i) $(2, 4)$.
 (iii) $(a - \varepsilon, a + \varepsilon)$.
 (v) $[-2, 2]$.
 (vii) $(-\infty, 1] \cup [1, \infty)$.
3. (i) All points below the graph of $f(x) = x$.
 (iii) All points below the graph of $f(x) = x^2$.
 (v) All points between the graphs of $f(x) = x + 1$ and $f(x) = x - 1$.
 (vii) A collection of straight lines parallel to the graph of $f(x) = -x$, intersecting the horizontal axis at the points $(n, 0)$ for integers n .
 (ix) All points inside the circle of radius 1 and around $(1, 2)$.
4. (i) A square with vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$.
 (iii) The union of the graph of $f(x) = x$ and of $f(x) = 2 - x$.
 (v) The point $(0, 0)$.
 (vii) The circle of radius $\sqrt{5}$ around $(1, 0)$, since $x^2 - 2x + y^2 = (x - 1)^2 + y^2 - 1$.
6. (a) Simply observe that the graph of $f(x) = m(x - a) + b = mx + (b - ma)$ is a straight line with slope m , which goes through the point (a, b) . (The important point about this exercise is simply to remember the point slope form.)
 (b) The straight line through (a, b) and (c, d) has slope $(d - b)/(c - a)$, so the equation follows from part (a).
 (c) When $m = m'$ and $b \neq b'$. In that case, there is clearly no number x with $f(x) = g(x)$, while such a number x always exists if $m \neq m'$, namely, $x = (b' - b)/(m - m')$.
7. (a) If $B = 0$ and $A \neq 0$, then the set is the vertical straight line formed by all points (x, y) with $x = -C/A$. If $B \neq 0$, the set is the graph of $f(x) = (-A/B)x + (-C/A)$.
 (b) The points (x, y) on the vertical line with $x = a$ are precisely the ones which satisfy $1 \cdot x + 0 \cdot y + (-a) = 0$. The points (x, y) on the graph of $f(x) = mx + b$ are precisely the ones which satisfy $(-m)x + 1 \cdot y + (-b) = 0$.
11. (i) The graph of f is symmetric with respect to the vertical axis.
 (ii) The graph of f is symmetric with respect to the origin. Equivalently, the part of the graph to the left of the vertical axis is obtained by reflecting first through the vertical axis, and then through the horizontal axis.
 (iii) The graph of f lies above or on the horizontal axis.
 (iv) The graph of f repeats the part between 0 and a over and over.
21. (a) The square of the distance from (x, x^2) to $(0, \frac{1}{4})$ is

$$\begin{aligned}
 x^2 + \left(x^2 - \frac{1}{4}\right)^2 &= x^2 + x^4 - \frac{x^2}{2} + \frac{1}{16} \\
 &= x^4 + \frac{x^2}{2} + \frac{1}{16} \\
 &= \left(x^2 + \frac{1}{4}\right)^2,
 \end{aligned}$$

which is the square of the distance from (x, x^2) to the graph of g .

- (b) The point (x, y) satisfies this condition if and only if

$$(x - \alpha)^2 + (y - \beta)^2 = (y - \gamma)^2,$$

or

$$x^2 - 2\alpha x + \alpha^2 + y^2 - 2\beta y + \beta^2 = y^2 - 2\gamma y + \gamma^2,$$

or

$$y = \left(\frac{1}{2\beta - 2\gamma} \right) x^2 + \left(\frac{\alpha}{\gamma - \beta} \right) x + \left(\frac{\alpha^2 + \beta^2 - \gamma^2}{2\beta - 2\gamma} \right).$$

If $\beta = \gamma$, so that P is on the line L , then the solution is the vertical line through P .

CHAPTER 5

1. (ii)

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} = \lim_{x \rightarrow 2} (x^2 + 2x + 4) = 12.$$

- (iv)

$$\begin{aligned} \lim_{x \rightarrow y} \frac{x^n - y^n}{x - y} &= \lim_{x \rightarrow y} x^{n-1} + x^{n-2}y + \cdots + xy^{n-2} + y^{n-1} \\ &= y^{n-1} + y^{n-1} + \cdots + y^{n-1} = ny^{n-1}. \end{aligned}$$

- (vi)

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h} &= \lim_{h \rightarrow 0} \frac{(\sqrt{a+h} - \sqrt{a})(\sqrt{a+h} + \sqrt{a})}{h(\sqrt{a+h} + \sqrt{a})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{a+h} + \sqrt{a}} \\ &= \frac{1}{2\sqrt{a}}. \end{aligned}$$

3. (i) $l = 0$. For all x we have $|\cos(x^2)| \leq 1$, so $|3 - \cos(x^2)| \leq 4$, and thus

$$|f(x) - 0| = |x| \cdot |3 - \cos(x^2)| \leq 4 \cdot |x|.$$

So we can take $\delta = \varepsilon/4$.

- (iii) $l = 100$. We have

$$\left| \frac{100}{x} - 100 \right| = 100 \cdot \left| \frac{1}{x} - 1 \right| = 100 \cdot \frac{1}{|x|} \cdot |x - 1|.$$

The initial stipulation $|x - 1| < \frac{1}{2}$ makes $x > \frac{1}{2}$, so $1/|x| < 2$, so we then have

$$|f(x) - 100| < 200 \cdot |x - 1|.$$

So we can take $\delta = \min(1/2, \varepsilon/200)$.

- (v) $l = 2$. The same sort of argument that was used in the text and in number (iii) shows that

$$\left| \frac{1}{x} - 1 \right| < \varepsilon \text{ for } 0 < |x - 1| < \delta_1 = \min(1/2, \varepsilon/2),$$

so that

$$\left| \frac{1}{x} - 1 \right| < \frac{\varepsilon}{2} \text{ for } 0 < |x - 1| < \bar{\delta}_1 = \min(1/2, \varepsilon/4).$$

Similarly, the solution to number (iv) gives a δ_2 such that

$$|x^4 - 1| < \varepsilon \text{ for } 0 < |x - 1| < \delta_2,$$

and we have a corresponding $\bar{\delta}_2$. Then we can take $\delta = \min(\bar{\delta}_1, \bar{\delta}_2)$.

- (vii) $l = 0$. Let $\delta = \varepsilon^2$.

6. (i) We need $|f(x) - 2| < \varepsilon/2$ and $|g(x) - 4| < \varepsilon/2$, so we need

$$0 < |x - 2| < \min\left(\sin^2\left(\frac{\varepsilon^2}{36}\right) + \frac{\varepsilon}{2}, \frac{\varepsilon^2}{4}\right) = \delta.$$

- (iii) We need

$$|g(x) - 4| < \min\left(\frac{|4|}{2}, \frac{\varepsilon|4|^2}{2}\right),$$

so we need

$$0 < |x - 2| < [\min(2, 8\varepsilon)]^2 = \delta.$$

9. Let $l = \lim_{x \rightarrow a} f(x)$ and define $g(h) = f(a + h)$. Then for every $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$. Now, if $0 < |h| < \delta$, then $0 < |(h + a) - a| < \delta$, so $|f(a + h) - l| < \varepsilon$. This inequality can be written $|g(h) - l| < \varepsilon$. Thus, $\lim_{h \rightarrow 0} g(h) = l$, which can also be written $\lim_{h \rightarrow 0} f(a + h) = l$. The same sort of argument shows that if $\lim_{h \rightarrow 0} f(a + h) = m$, then $\lim_{x \rightarrow a} f(x) = m$. So either limit exists if the other does, and in this case they are equal.
10. (a) Intuitively, we can get $f(x)$ as close to l as we like if and only if we can get $f(x) - l$ as close to 0 as we like. The formal proof is so trivial that it takes a bit of work to make it look like a proof at all. To be very precise, suppose $\lim_{x \rightarrow a} f(x) = l$ and let $g(x) = f(x) - l$. Then for all $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$. This last inequality can be written $|g(x) - 0| < \varepsilon$, so $\lim_{x \rightarrow a} g(x) = 0$. The argument in the other direction is similarly uninteresting.
- (b) Intuitively, making x close to a is the same as making $x - a$ close to 0. Formally: Suppose that $\lim_{x \rightarrow a} f(x) = l$, and let $g(x) = f(x - a)$. Then for all $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$. Now, if $0 < |y| < \delta$, then $0 < |(y + a) - a| < \delta$, so

$|f(y+a) - l| < \varepsilon$. But this last inequality can be written $|g(y) - l| < \varepsilon$. So $\lim_{y \rightarrow 0} g(y) = l$. The argument in the reverse direction is similar.

- (c) Intuitively, x is close to 0 if and only if x^3 is. Formally: Let $\lim_{x \rightarrow 0} f(x) = l$. For every $\varepsilon > 0$ there is a $\delta > 0$ such that if $0 < |x| < \delta$, then $|f(x) - l| < \varepsilon$. Then if $0 < |x| < \min(1, \delta)$, we have $0 < |x^3| < \delta$, so $|f(x^3) - l| < \varepsilon$. Thus, $\lim_{x \rightarrow 0} f(x^3) = l$. On the other hand, if we assume that $\lim_{x \rightarrow 0} f(x^3)$ exists, say $\lim_{x \rightarrow 0} f(x^3) = m$, then for all $\varepsilon > 0$ there is a δ such that if $0 < |x| < \delta$, then $|f(x^3) - m| < \varepsilon$. Then if $0 < |x| < \delta^3$, we have $0 < |\sqrt[3]{x}| < \delta$, so $|f([\sqrt[3]{x}]^3) - m| < \varepsilon$, or $|f(x) - m| < \varepsilon$. Thus $\lim_{x \rightarrow 0} f(x) = m$.
- (d) Let $f(x) = 1$ for $x \geq 0$, and $f(x) = -1$ for $x < 0$. Then $\lim_{x \rightarrow 0} f(x^2) = 1$, but $\lim_{x \rightarrow 0} f(x)$ does not exist.
17. (a) The function $f(x) = 1/x$ cannot approach a limit at 0, since it becomes arbitrarily large near 0. In fact, no matter what $\delta > 0$ may be, there is some x satisfying $0 < |x| < \delta$, but $1/x > |l| + \varepsilon$, namely, any $x < \min(\delta, 1/(|l| + \varepsilon))$. Any such x does not satisfy $|1/x - l| < \varepsilon$.
- (b) No matter what $\delta > 0$ may be, there is some x satisfying $0 < |x - 1| < \delta$, but $1/(x - 1) > |l| + \varepsilon$, namely, any $x < \min(1 + \delta, 1/(|l| + \varepsilon))$. Such an x does not satisfy $|1/(x - 1) - l| < \varepsilon$. (It is also possible to apply Problem 10(b): $\lim_{x \rightarrow 0} 1/x = \lim_{x \rightarrow 1} 1/(x - 1)$ if the latter exists, so this limit does not exist, because of part (a).)
25. (i) This is the usual definition, simply calling the numbers δ and ε , instead of ε and δ .
- (ii) This is a minor modification of (i): if the condition is true for *all* $\delta > 0$, then it applies to $\delta/2$, so there is an $\varepsilon > 0$ such that if $0 < |x - a| < \varepsilon$, then $|f(x) - l| \leq \delta/2 < \delta$.
- (iii) This is a similar modification: apply it to $\delta/5$ to obtain (i).
- (iv) This is also a modification: it says the same thing as (i), since $\varepsilon/10 > 0$, and it is only the existence of *some* $\varepsilon > 0$ that is in question.
29. If $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = l$, then for every $\varepsilon > 0$ there are $\delta_1, \delta_2 > 0$ such that, for all x ,

$$\begin{aligned} \text{if } a < x < a + \delta_1, \text{ then } |f(x) - l| < \varepsilon, \\ \text{if } a - \delta_2 < x < a, \text{ then } |f(x) - l| < \varepsilon. \end{aligned}$$

Let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then either $a - \delta_2 < a - \delta < x < a$ or else $a < x < a + \delta < a + \delta_1$, so $|f(x) - l| < \varepsilon$.

30. (i) If $l = \lim_{x \rightarrow 0^+} f(x)$, then for all $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - l| < \varepsilon$ for $0 < x < \delta$. If $-\delta < x < 0$, then $0 < -x < \delta$, so $|f(-x) - l| < \varepsilon$. Thus $\lim_{x \rightarrow 0^-} f(-x) = l$. Similarly, if $\lim_{x \rightarrow 0^-} f(x)$ exists, then $\lim_{x \rightarrow 0^+} f(x)$ exists and has the same value. (Intuitively, x is close to 0 and positive if and only if $-x$ is close to 0 and negative.)

- (ii) If $l = \lim_{x \rightarrow 0^+} f(x)$, then for all $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - l| < \varepsilon$ for $0 < x < \delta$. So if $0 < |x| < \delta$, then $|f(|x|) - l| < \varepsilon$. Thus $\lim_{x \rightarrow 0} f(|x|) = l$. The reverse direction is similar. (Intuitively, if x is close to 0, then $|x|$ is close to 0 and positive.)
- (iii) If $l = \lim_{x \rightarrow 0^+} f(x)$, then for all $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - l| < \varepsilon$ for $0 < x < \delta$. If $0 < |x| < \sqrt{\delta}$, then $0 < x^2 < \delta$, so $|f(x^2) - l| < \varepsilon$. Thus $\lim_{x \rightarrow 0} f(x^2) = l$. The reverse direction is similar. (Intuitively, if x is close to 0, then x^2 is close to 0 and positive.)
34. If $l = \lim_{x \rightarrow \infty} f(x)$, then for every $\varepsilon > 0$ there is some N such that $|f(x) - l| < \varepsilon$ for $x > N$, and we can clearly assume that $N > 0$. Now, if $0 < x < 1/N$, then $1/x > N$, so $|f(1/x) - l| < \varepsilon$. Thus $\lim_{x \rightarrow 0^+} f(1/x) = l$. The reverse direction is similar.

CHAPTER 6

1. (i) $F(x) = x + 2$ for all x .
- (iii) $F(x) = 0$ for all x .

CHAPTER 7

1. (i) Bounded above and below; minimum value 0; no maximum value.
- (iii) Bounded below but not above; minimum value 0.
- (v) Bounded above and below. If $a \leq -1/2$, then $a \leq -a - 1$, so $f(x) = a + 2$ for all x in $(-a - 1, a + 1)$, so $a + 2$ is the maximum and minimum value. If $-1/2 < a \leq 0$, then f has the minimum value a^2 , and if $a \geq 0$, then f has the minimum value 0. Since $a + 2 > (a + 1)^2$ only for $(-1 - \sqrt{5})/2 < a < (-1 + \sqrt{5})/2$, when $a \geq -1/2$ the function f has a maximum value only for $a \leq (-1 + \sqrt{5})/2$ (the maximum value being $a + 2$).
- (vii) Bounded above and below; maximum value 1; minimum value 0.
- (ix) Bounded above and below; maximum value 1; minimum value -1 .
- (xi) f has a maximum and minimum value, since f is continuous.
2. (i) $n = -2$, since $f(-2) < 0 < f(-1)$.
- (iii) $n = -1$, since $f(-1) = -1 < 0 < f(0)$.
3. (i) If $f(x) = x^{179} + 163/(1 + x^2 + \sin^2 x) - 119$, then f is continuous on $\mathbf{R}$ and $f(2) > 0$, while $f(-2) < 0$, so $f(x) = 0$ for some x in $(-2, 2)$.
5. f is constant, for if f took on two different values, then f would take on all values in between, which would include irrational values.
7. (1) $f(x) = x$;
(2) $f(x) = -x$;
(3) $f(x) = |x|$;
(4) $f(x) = -|x|$.
10. Apply Theorem 1 to $f - g$.

11. If $f(0) = 0$ or $f(1) = 1$, choose $x = 0$ or 1 . If $f(0) > 0 = I(0)$ and $f(1) < 1 = I(1)$, then Problem 10 applied to f and I implies that $f(x) = x$ for some x .

CHAPTER 8

1. (i) 1 is the greatest element, and the greatest lower bound is 0, which is not in the set.
 (iii) 1 is the greatest element, and 0 is the least element.
 (v) Since $\{x : x^2 + x + 1 \geq 0\} = \mathbf{R}$, there is no least upper bound or greatest lower bound.
 (vii) Since $\{x : x < 0 \text{ and } x^2 + x - 1 < 0\} = ([-1 - \sqrt{5}]/2, 0)$, the greatest lower bound is $[-1 - \sqrt{5}]/2$, and the least upper bound is 0; neither belongs to the set.
2. (a) Since $A \neq \emptyset$, there is some x in A . Then $-x$ is in $-A$, so $-A \neq \emptyset$. Since A is bounded below, there is some y such that $y \leq x$ for all x in A . Then $-y \geq -x$ for all x in A , so $-y \geq z$ for all z in $-A$, so $-A$ is bounded above. Let $\alpha = \sup(-A)$. Then α is an upper bound for $-A$, so, reversing the argument just given, $-\alpha$ is a lower bound for A .
 Moreover, if β is any lower bound for A , then $-\beta$ is an upper bound for $-A$, so $-\beta \geq \alpha$, so $\beta \leq -\alpha$. Thus $-\alpha$ is the greatest lower bound for A .
5. (a) If l is the largest integer with $l \leq x$, then $l + 1 > x$, but $l + 1 \leq x + 1 < y$. So we can let $k = l + 1$. (Proof that a largest such integer l exists: Since $\mathbf{N}$ is not bounded above, there is some natural number n with $-n < x < n$. There are consequently only a finite number of integers l with $-n \leq l \leq x$. Pick the largest.)
 (b) Since $y - x > 0$, there is some natural number n with $1/n < y - x$. Since $ny - nx > 1$, there is, by part (a), an integer k with $nx < k < ny$, which means that $x < k/n < y$.
 (c) Choose $r + \sqrt{2}(s - r)/2$.
 (d) By part (b), there is a rational number r with $x < r < y$, and therefore a rational number s with $x < r < s < y$. Apply part (c) to $r < s$.
10. Let k be the largest integer $\leq x/\alpha$ (the solution to Problem 5 shows that such a k exists), and let $x' = x - k\alpha \geq 0$. If $x - k\alpha = x' \geq \alpha$, then $x \geq (k + 1)\alpha$, so $k + 1 \leq x/\alpha$, contradicting the choice of k . So $0 \leq x' < \alpha$.
12. (a) Since any y in B satisfies $y \geq x$ for all x in A , any y in B is an upper bound for A , so $y \geq \sup A$.
 (b) Part (a) shows that $\sup A$ is a lower bound for B , so $\sup A \leq \inf B$.
13. Since $x \leq \sup A$ and $y \leq \sup B$ for every x in A , and y in B , it follows that $x + y \leq \sup A + \sup B$. Thus, $\sup A + \sup B$ is an upper bound for $A + B$, so $\sup(A + B) \leq \sup A + \sup B$. If x and y are chosen in A and B , respectively, so that $\sup A - x < \varepsilon/2$ and $\sup B - y < \varepsilon/2$, then $\sup A + \sup B - (x + y) < \varepsilon$. Hence,

$$\sup(A + B) \geq x + y > \sup A + \sup B - \varepsilon.$$

CHAPTER 9

1. (a)

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1}{a(a+h)} = -\frac{1}{a^2}. \end{aligned}$$

(b) The tangent line through $(a, 1/a)$ is the graph of

$$\begin{aligned} g(x) &= \frac{-1}{a^2}(x-a) + \frac{1}{a} \\ &= \frac{-x}{a^2} + \frac{2}{a}. \end{aligned}$$

If $f(x) = g(x)$, then

$$\frac{1}{x} = -\frac{x}{a^2} + \frac{2}{a}$$

or

$$x^2 - 2ax + a^2 = 0,$$

so $x = a$.

2. (a)

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{(a+h)^2} - \frac{1}{a^2}}{h} \\ &= \lim_{h \rightarrow 0} \frac{(-2ah - h^2)}{ha^2(a+h)^2} = -\frac{2}{a^3}. \end{aligned}$$

(b) The tangent line through $(a, 1/a^2)$ is the graph of

$$\begin{aligned} g(x) &= -\frac{2}{a^3}(x-a) + \frac{1}{a^2} \\ &= -\frac{2x}{a^3} + \frac{3}{a^2}. \end{aligned}$$

If $f(x) = g(x)$, then

$$\frac{1}{x^2} = -\frac{2x}{a^3} + \frac{3}{a^2},$$

or

$$2x^3 - 3ax^2 + a^3 = 0,$$

or

$$0 = (x-a)(2x^2 - ax - a^2) = (x-a)(2x+a)(x-a).$$

So $x = a$ or $x = -a/2$; the point $(-a/2, 4/a^2)$ lies on the opposite side of the vertical axis from $(a, 1/a^2)$.

3.

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h} \\ &= \lim_{h \rightarrow 0} \frac{(\sqrt{a+h} - \sqrt{a})(\sqrt{a+h} + \sqrt{a})}{h(\sqrt{a+h} + \sqrt{a})} = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{a+h} + \sqrt{a})} \\ &= \frac{1}{2\sqrt{a}}. \end{aligned}$$

 4. Conjecture: $S_n'(x) = nx^{n-1}$. Proof:

$$\begin{aligned} S_n'(x) &= \lim_{h \rightarrow 0} \frac{S_n(x+h) - S_n(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sum_{j=0}^n \binom{n}{j} x^{n-j} h^j - x^n}{h} \\ &= \lim_{h \rightarrow 0} \sum_{j=1}^n \binom{n}{j} x^{n-j} h^{j-1} \\ &= \binom{n}{1} x^{n-1} = nx^{n-1}, \quad \text{since } \lim_{h \rightarrow 0} h^{j-1} = 0 \text{ for } j > 1. \end{aligned}$$

 5. $f'(x) = 0$ for x not an integer, and $f'(x)$ is not defined if x is an integer.

6. (a)

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{[f(x+h) + c] - [f(x) + c]}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x). \end{aligned}$$

(b)

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\ &= c \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = cf'(x). \end{aligned}$$

 7. (a) $f'(9) = 3 \cdot 9^2$; $f'(25) = 3 \cdot (25)^2$; $f'(36) = 3 \cdot (36)^2$.

 (b) $f'(3^2) = f'(9) = 3 \cdot 9^2$; $f'(5^2) = f'(25) = 3 \cdot (25)^2$; $f'(6^2) = f'(36) = 3 \cdot (36)^2$.

 (c) $f'(a^2) = 3(a^2)^2 = 3a^4$; $f'(x^2) = 3(x^2)^2 = 3x^4$.

 (d) $f'(x^2) = 3x^4$, but $g(x) = x^6$, so $g'(x) = 6x^5$.

8. (a)

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x+h+c) - f(x+c)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f([x+c]+h) - f(x+c)}{h} = f'(x+c). \end{aligned}$$

(b)

$$\begin{aligned}
 g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{f(cx+ch) - f(cx)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{c[f(cx+ch) - f(cx)]}{ch} = \lim_{k \rightarrow 0} \frac{c[f(cx+k) - f(cx)]}{k} \\
 &= c \cdot \lim_{k \rightarrow 0} \frac{f(cx+k) - f(cx)}{k} = c \cdot f'(cx).
 \end{aligned}$$

(Compare the manipulations in this calculation with Problem 5-14.)

- (c) If $g(x) = f(x+a)$, then $g'(x) = f'(x+a)$, by part (a). But $g = f$, so $f'(x) = g'(x) = f'(x+a)$ for all x , which means that f' is periodic, with period a .
9. (i) If $g(x) = x^5$, then $g'(x) = 5x^4$. Now $f(x) = g(x+3)$, so by Problem 8(a), $f'(x) = g'(x+3) = 5(x+3)^4$. And $f'(x+3) = 5(x+6)^4$.
 (ii) $f(x) = (x-3)^5$, so $f'(x) = 5(x-3)^4$, as in part (i). And $f'(x+3) = 5x^4$.
 (iii) $f(x) = (x+2)^7$, so $f'(x) = 7(x+2)^6$, as in part (i). And $f'(x+3) = 7(x+5)^6$.
10. If $f(x) = g(t+x)$, then $f'(x) = g'(t+x)$, by Problem 8(a). If $f(t) = g(t+x)$, then $f'(t) = g'(t+x)$, by Problem 8(a), so $f'(x) = g'(2x)$.
11. (a) If $s(t) = ct^2$, then $s'(t) = 2ct$, and there is no number k such that $s'(t) = ks(t)$ [that is, $2ct = kct^2$] for all t .
 (By the way, at this point we do not know any nonzero function f for which f' is proportional to f . After Chapter 18 it might be amusing to determine what the world would be like if Galileo were correct.)
- (b) (i) If $s(t) = (a/2)t^2$, then $s'(t) = at$, so $s''(t) = a$.
 (ii) $[s'(t)]^2 = (at)^2 = 2a \cdot (a/2)t^2 = 2as(t)$.
- (c) The chandelier falls $s(t) = 16t^2$ feet in t seconds, so it falls 400 feet in t seconds, if $400 = 16t^2$, or $t = 5$. After 5 seconds the velocity will be $s'(5) = 5a = 5 \cdot 32 = 160$ feet per second. The speed was half this amount when $80 = s'(t) = 32t$, or $t = \frac{5}{2}$.
21. (a) This is another way of writing the definition (see Problem 5-9).
 (b) This follows from Problem 5-11, applied to the functions $\alpha(h) = [f(a+h) - f(a)]/h$ and $\beta(h) = [g(a+h) - g(a)]/h$.
26. (i) $f''(x) = 6x$.
 (iii) $f''(x) = 4x^3$.
30. (i) means that $f'(a) = na^{n-1}$ if $f(x) = x^n$.
 (iii) means that $g'(a) = f'(a)$ if $g(x) = f(x) + c$.
 (v) means the same as (iii).
 (vii) means that $g'(b) = f'(b+a)$ if $g(x) = f(x+a)$.
 (ix) means that $g'(b) = cf'(cb)$ if $g(x) = f(cx)$.

CHAPTER 10

1. (i) $(1+2x) \cdot \cos(x+x^2)$.
 (iii) $(-\sin x) \cdot \cos(\cos x)$.
 (v) $\cos\left(\frac{\cos x}{x}\right) \cdot \frac{-x \sin x - \cos x}{x^2}$.

- (vii) $(\cos(x + \sin x)) \cdot (1 + \cos x)$.
2. (i) $(\cos((x + 1)^2(x + 2))) \cdot [2(x + 1)(x + 2) + (x + 1)^2]$.
- (iii) $[2 \sin((x + \sin x)^2) \cos((x + \sin x)^2)] \cdot 2(x + \sin x)(1 + \cos x)$.
- (v) $(\cos(x \sin x)) \cdot (\sin x + x \cos x) + (\cos(\sin x^2)(\cos x^2)) \cdot 2x$.
- (vii) $(2 \sin x \cos x \sin x^2 \sin^2 x^2) + (2x \cos x^2 \sin^2 x \sin^2 x^2) + (4x \sin x^2 \cos x^2 \sin^2 x \sin x^2)$.
- (ix) $6(x + \sin^5 x)^5(1 + 5 \sin^4 x \cos x)$.
- (xi) $\cos(\sin^7 x^7 + 1)^7 \cdot 7(\sin^7 x^7 + 1)^6 \cdot (7 \sin^6 x^7 \cdot \cos x^7 \cdot 7x^6)$.
- (xiii) $\cos(x^2 + \sin(x^2 + \sin x^2)) \cdot [(2x + \cos(x^2 + \sin x^2)) \cdot (2x + 2x \cos x^2)]$.
- (xv) $\frac{(1 + \sin x)(2x \cos x^2 \cdot \sin^2 x + \sin x^2 \cdot 2 \sin x \cos x) - \cos x \sin x^2 \sin^2 x}{(1 + \sin x)^2}$.

$$(xvii) \cos \left(\frac{x^3}{\sin \left(\frac{x^3}{\sin x} \right)} \right)$$

$$\frac{3x^2 \sin \left(\frac{x^3}{\sin x} \right) - x^3 \cos \left(\frac{x^3}{\sin x} \right) \cdot \left(\frac{3x^2 \sin x - x^3 \cos x}{\sin^2 x} \right)}{\sin^2 \left(\frac{x^3}{\sin x} \right)}$$

4. (i) $-\frac{(x + 1)^2}{(x + 2)^2}$.
- (iii) $2x^2$.
5. (i) $-x^2$.
- (iii) 17.
6. (i) $f'(x) = g'(x + g(a))$.
- (iii) $f'(x) = g'(x + g(x)) \cdot (1 + g'(x))$.
- (v) $f'(x) = g(a)$.
7. (a) $A'(t) = 2\pi r(t)r'(t)$. Since $r'(t) = 4$ for that t with $r(t) = 6$, it follows that $A'(t) = 2\pi \cdot 6 \cdot 4 = 48\pi$ when $r(t) = 6$.
- (b) If $V(t)$ is the volume at time t , then $V(t) = 4\pi r(t)^3/3$, so $V'(t) = 4\pi r(t)^2 r'(t) = 4\pi \cdot 6^2 \cdot 4 = 576\pi$ when $r(t) = 6$.
- (c) First method: Since $A'(t) = 2\pi r(t)r'(t)$, and $A'(t) = 5$ for $r(t) = 3$, it follows that

$$r'(t) = \frac{A'(t)}{2\pi r(t)} = \frac{5}{6\pi} \quad \text{when } r(t) = 3.$$

Thus

$$\begin{aligned} V'(t) &= 4\pi r(t)^2 r'(t) \\ &= 4\pi \cdot 9 \cdot \frac{5}{6\pi} \\ &= 30 \quad \text{when } r(t) = 3. \end{aligned}$$

To apply the second method, we first note that if

$$f(t) = A(t)^{3/2} = \sqrt{A(t)^3},$$

then, using Problem 9-3 and the Chain Rule,

$$\begin{aligned} f'(t) &= \frac{1}{2\sqrt{A(t)^3}} \cdot 3A(t)^2 A'(t) \\ &= \frac{1}{2A(t)^{3/2}} \cdot 3A(t)^2 A'(t) \\ &= \frac{3}{2} A(t)^{1/2} A'(t) \quad (\text{just as we might have guessed}). \end{aligned}$$

Now

$$\begin{aligned} V(t) &= \frac{4\pi r(t)^3}{3} = \frac{4\pi [r(t)^2]^{3/2}}{3} \\ &= \frac{4[\pi r(t)^2]^{3/2}}{3\pi^{1/2}} \\ &= \frac{4A(t)^{3/2}}{3\pi^{1/2}}. \end{aligned}$$

So

$$\begin{aligned} V'(t) &= \frac{4}{3\pi^{1/2}} \cdot \frac{3}{2} \sqrt{A(t)} A'(t) \\ &= \frac{2}{\pi^{1/2}} \cdot \pi^{1/2} r(t) A'(t) \\ &= 2 \cdot 3 \cdot 5 = 30. \end{aligned}$$

10. (i) $(f \circ h)'(0) = f'(h(0)) \cdot h'(0) = f'(3) \cdot \sin^2(\sin 1) = [6 \sin \frac{1}{3} - \cos \frac{1}{3}] \sin^2(\sin 1).$
 (iii) $\alpha'(x^2) = h'(x^4) \cdot 2x^2 = \sin^2(\sin(x^4 + 1)) \cdot 2x^2.$
 12. The Chain Rule implies that

$$\begin{aligned} \left(\frac{1}{g}\right)'(x) &= (f \circ g)'(x) = f'(g(x)) \cdot g'(x) \\ &= -\frac{1}{g(x)^2} \cdot g'(x). \end{aligned}$$

35. (i) $\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx} = (\cos y) \cdot (1 + 2x) = (\cos(x + x^2)) \cdot (1 + 2x).$
 (iii) $\frac{dz}{dx} = \frac{dz}{du} \cdot \frac{du}{dx} = (\cos u) \cdot (\cos x) = (\cos(\sin x)) \cdot (\cos x).$

CHAPTER 11

1. (i) $0 = f'(x) = 3x^2 - 2x - 8$ for $x = 2$ and $x = -\frac{4}{3}$, both of which are in $[-2, 2]$;
 $f(-2) = 5$, $f(2) = -11$, $f(-\frac{4}{3}) = \frac{203}{27}$;
 maximum = $\frac{203}{27}$, minimum = -11 .

- (iii) $0 = f'(x) = 12x^3 - 24x^2 + 12x = 12x(x^2 - 2x + 1)$ for $x = 0$ and $x = 1$, of which only 0 is in $[-\frac{1}{2}, \frac{1}{2}]$;
 $f(-\frac{1}{2}) = \frac{43}{16}$, $f(\frac{1}{2}) = \frac{11}{16}$, $f(0) = 0$;
 maximum = $\frac{43}{16}$, minimum = 0.

- (v) $0 = f'(x) =$

$$\frac{x^2 + 1 - (x + 1)2x}{(x^2 + 1)^2} = \frac{1 - 2x - x^2}{(x^2 + 1)^2}$$

for $x = -1 + \sqrt{2}$ and $x = -1 - \sqrt{2}$, of which only $-1 + \sqrt{2}$ is in $[-1, \frac{1}{2}]$;

$$f(-1) = 0, f(\frac{1}{2}) = \frac{6}{5}, f(-1 + \sqrt{2}) = (1 + \sqrt{2})/2;$$

maximum = $(1 + \sqrt{2})/2$, minimum = 0.

2. (i) $-\frac{4}{3}$ is a local maximum point, and 2 is a local minimum point.
 (iii) 0 is a local minimum point, and there are no local maximum points.
 (v) $-1 + \sqrt{2}$ is a local maximum point, and $-1 - \sqrt{2}$ is a local minimum point.
4. (a) Notice that f actually has a minimum value, since f is a polynomial function of even degree. The minimum occurs at a point x with

$$0 = f'(x) = 2 \sum_{i=1}^n (x - a_i),$$

so $x = (a_1 + \cdots + a_n)/n$.

5. (i) 3 and 7 are local maximum points, and 5 and 9 are local minimum points.
 (iii) All irrational $x > 0$ are local minimum points, and all irrational $x < 0$ are local maximum points.
 (v) x is a local minimum point if its decimal expansion does not contain a 5. It is a local maximum point if its decimal expansion contains exactly one 5 that is followed by an infinite string of 9's. In all other cases, x is both a local maximum point and a local minimum point.
7. If $f(x)$ is the total length of the path, then

$$f(x) = \sqrt{x^2 + a^2} + \sqrt{(1-x)^2 + b^2}.$$

The positive function f clearly has a minimum, since $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = \infty$, and f is differentiable everywhere, so the minimum occurs at a point x with $f'(x) = 0$. Now, $f'(x) = 0$ when

$$\frac{x}{\sqrt{x^2 + a^2}} - \frac{(1-x)}{\sqrt{(1-x)^2 + b^2}} = 0.$$

This equation says that $\cos \alpha = \cos \beta$.

It is also possible to notice that $f(x)$ is equal to the sum of the lengths of the dashed line segment and the line segment from $(x, 0)$ to $(1, b)$. This is shortest when the two line segments lie along a line (because of Problem 4-9(b), if

a rigorous reason is required); a little plane geometry shows that this happens when $\alpha = \beta$.

10. If x is the length of one side of a rectangle of perimeter P , then the length of the other side is $(P - 2x)/2$, so the area is

$$A(x) = \frac{x(P - 2x)}{2}.$$

So the rectangle with greatest area occurs when x is the maximum point for f on $(0, P/2)$. Since A is continuous on $[0, P/2]$, and $A(0) = A(P/2) = 0$, and $A(x) > 0$ for x in $(0, P/2)$, the maximum exists. Since A is differentiable on $(0, P/2)$, the minimum point x satisfies

$$\begin{aligned} 0 = A'(x) &= \frac{P - 2x}{2} - x \\ &= \frac{P - 4x}{2}, \end{aligned}$$

so $x = P/4$.

11. Let $S(r)$ be the surface area of the right circular cylinder of volume V with radius r . Since

$$V = \pi r^2 h \quad \text{where } h \text{ is the height,}$$

we have $h = V/\pi r^2$, so

$$\begin{aligned} S(r) &= 2\pi r^2 + 2\pi r h \\ &= 2\pi r^2 + \frac{2V}{r}. \end{aligned}$$

We want the minimum point of S on $(0, \infty)$; this exists, since $\lim_{r \rightarrow 0} S(r) = \lim_{r \rightarrow \infty} S(r) = \infty$. Since S is differentiable on $(0, \infty)$, the minimum point r satisfies

$$\begin{aligned} 0 = S'(r) &= 4\pi r - \frac{2V}{r^2} \\ &= \frac{4\pi r^3 - 2V}{r^2}, \end{aligned}$$

or

$$r = \sqrt[3]{\frac{V}{2\pi}}.$$

21. 1 is a local maximum point, and 3 is a local minimum point.
28. (a) We have

$$\begin{aligned} \frac{f(b) - f(a)}{b - a} &= f'(x) \quad \text{for some } x \text{ in } (a, b) \\ &\geq M, \end{aligned}$$

$$\text{so } f(b) - f(a) \geq M(b - a).$$

(b) We have

$$\frac{f(b) - f(a)}{b - a} = f'(x) \quad \text{for some } x \text{ in } (a, b)$$

$$\leq m,$$

so $f(b) - f(a) \leq m(b - a)$.

(c) If $|f'(x)| \leq M$ for all x in $[a, b]$, then $-M \leq f'(x) \leq M$, so

$$f(a) - M(b - a) \leq f(b) \leq f(a) + M(b - a),$$

or

$$|f(b) - f(a)| \leq M(b - a).$$

31. (a) $f(x) = -\cos x + a$ for some number a (because $f(x) = -\cos x$ is one such function, and any two such functions differ by a constant function).
 (b) $f'(x) = x^4/4 + a$ for some number a , so $f(x) = x^5/20 + ax + b$ for some numbers a and b .
 (c) $f''(x) = x^2 + x^3/3 + a$ for some a , so $f'(x) = x^3/6 + x^4/12 + ax + b$ for some a and b , so $f(x) = x^4/24 + x^5/60 + ax^2/2 + bx + c$ for some numbers a , b , and c . Equivalently, and more simply, $f(x) = x^4/24 + x^5/60 + ax^2 + bx + c$ for some numbers a , b , and c .
32. (a) Since $s''(t) = -32$, we have $s'(t) = -32t + \alpha$ for some α , so $s(t) = -16t^2 + \alpha t + \beta$ for some α and β .
 (b) Clearly, $s(0) = 0 + 0 + \beta = 0 + \alpha$. Thus, $\alpha = v_0$ and $\beta = s_0$.
 (c) In this case, $s_0 = 0$ and $v_0 = v$, so $s(t) = -16t^2 + vt$. The maximum value of s occurs when $0 = s'(t) = -32t + v$, or $t = v/32$, so the maximum value is

$$s\left(\frac{v}{32}\right) = -16\left(\frac{v}{32}\right)^2 + v \cdot \left(\frac{v}{32}\right)$$

$$= \frac{-v^2}{64} + \frac{v^2}{32}$$

$$= \frac{v^2}{64}.$$

At that moment the velocity is clearly 0, but the acceleration is -32 (as at any time). The weight hits the ground at time $t > 0$ when

$$0 = s(t) = -16t^2 + vt,$$

or $t = v/16$ (it takes as long to fall back down as it took to reach the top). The velocity is then

$$s'(v/16) = -32\left(\frac{v}{16}\right) + v$$

$$= -v$$

(the same velocity with which it was initially moving upward).

47. Apply the Mean Value Theorem to $f(x) = \sqrt{x}$ on $[64, 66]$:

$$\frac{\sqrt{66} - \sqrt{64}}{66 - 64} = f'(x) = \frac{1}{2\sqrt{x}} \quad \text{for some } x \text{ in } [64, 66].$$

Since $64 < x < 81$, we have $8 < \sqrt{x} < 9$, so

$$\frac{1}{2 \cdot 9} < \frac{\sqrt{66} - 8}{2} < \frac{1}{2 \cdot 8}.$$

51. l'Hôpital's Rule does not lead to the equation

$$\lim_{x \rightarrow 1} \frac{3x^2 + 1}{2x - 3} = \lim_{x \rightarrow 1} \frac{6x}{2}$$

because $\lim_{x \rightarrow 1} 3x^2 + 1 \neq 0$.

52. (i)

$$\lim_{x \rightarrow 0} \frac{x}{\tan x} = \lim_{x \rightarrow 0} \frac{1}{\sec^2 x} = \lim_{x \rightarrow 0} \cos^2 x = 1.$$

(ii)

$$\lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x^2} = \lim_{x \rightarrow 0} \frac{-2 \sin x \cos x}{2x} = -1.$$

CHAPTER 12

1. (i) $f^{-1}(x) = (x - 1)^{1/3}$. (If $y = f^{-1}(x)$, then $x = f(y) = y^3 + 1$, so $y = (x - 1)^{1/3}$.)
 (iii) $f^{-1} = f$. (If $y = f^{-1}(x)$, then

$$x = f(y) = \begin{cases} y, & y \text{ rational} \\ -y, & y \text{ irrational;} \end{cases}$$

since $\pm y$ is rational or irrational if and only if y is, we have $y = x$ if x is rational and $y = -x$ if x is irrational, so $y = f(x)$.)

(v)

$$f^{-1}(x) = \begin{cases} x, & x \neq a_1, \dots, a_n \\ a_{i-1}, & x = a_i, \quad i = 2, \dots, n \\ a_n, & x = a_1. \end{cases}$$

(vii) $f^{-1} = f$.

2. (i) f^{-1} is increasing and $f^{-1}(x)$ is not defined for $x \leq 0$.
 (iii) f^{-1} is decreasing and $f^{-1}(x)$ is not defined for $x \leq 0$.
 3. Suppose f is increasing. Let $a < b$. Then $f^{-1}(a) \neq f^{-1}(b)$, since f^{-1} is one-one. So either $f^{-1}(a) < f^{-1}(b)$ or $f^{-1}(a) > f^{-1}(b)$. But if $f^{-1}(a) > f^{-1}(b)$, then

$$b = f(f^{-1}(b)) < f(f^{-1}(a)) = a,$$

a contradiction. The proof is similar for decreasing f , or one can consider $-f$ instead.

4. Clearly, $f + g$ is increasing, for if $f(a) < f(b)$ and $g(a) < g(b)$, then $(f + g)(a) = f(a) + g(a) < f(b) + g(b) = (f + g)(b)$.
 $f \cdot g$ is not necessarily increasing; for example, if $f(x) = g(x) = x$. (But $f \cdot g$

is increasing if $f(x), g(x) \geq 0$ for all x .)

$f \circ g$ is increasing, for if $a < b$, then $g(a) < g(b)$, so $f(g(a)) < f(g(b))$.

5. (a) If $(f \circ g)(x) = (f \circ g)(y)$, so that $f(g(x)) = f(g(y))$, then $g(x) = g(y)$, since f is one-one, so $x = y$, since g is one-one.
 $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$: for if $y = (f \circ g)^{-1}(x)$, then $x = (f \circ g)(y) = f(g(y))$, so $g(y) = f^{-1}(x)$, so $y = g^{-1}(f^{-1}(x))$.
6. If $f(x) = f(y)$, then

$$\frac{ax + b}{cx + d} = \frac{ay + b}{cy + d},$$

so

$$acxy + bcy + adx + bd = acxy + ady + bcx + bd,$$

or

$$ad(x - y) = bc(x - y).$$

If $ad \neq bc$, this implies that $x - y = 0$. (But if $ad = bc$, then $f(x) = f(y)$ for all x and y in the domain of f .)

If $y = f^{-1}(x)$, then $x = f(y)$, so

$$x = \frac{ay + b}{cy + d}$$

so

$$f^{-1}(x) = y = \frac{-dx + b}{cx - a} \quad \text{for } x \neq a/c.$$

7. (i) Those intervals $[a, b]$ which are contained in $(-\infty, 0]$ or $[0, 2]$ or $[2, \infty)$, since f is increasing on $(-\infty, 0]$ and $[2, \infty)$, and decreasing on $[0, 2]$.
 (iii) Those intervals $[a, b]$ which are contained in $(-\infty, 0]$ or $[0, \infty)$, since f is increasing on $(-\infty, 0]$ and decreasing on $[0, \infty)$.

11. Since

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))},$$

we have

$$\begin{aligned} (f^{-1})''(x) &= \frac{-f''(f^{-1}(x)) \cdot (f^{-1})'(x)}{[f'(f^{-1}(x))]^2} \\ &= \frac{-f''(f^{-1}(x))}{[f'(f^{-1}(x))]^3}. \end{aligned}$$

20. The formula for the derivative reads:

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}.$$

(In this formula, it is understood that dx/dy means $(f^{-1})'(y)$, while dy/dx is an “expression involving x ,” and in the final answer x must be replaced by y , by means of the equation $y = f(x)$.)

The computation in Problem 20, when completed, shows that

$$\begin{aligned}\frac{dx^{1/n}}{dx} &= \frac{1}{n(x^{1/n})^{n-1}} = \frac{1}{nx^{1-(1/n)}} \\ &= \frac{1}{n}x^{(1/n)-1}.\end{aligned}$$

21.

$$\begin{aligned}G'(x) &= x(f^{-1})'(x) + f^{-1}(x) - F'(f^{-1}(x)) \cdot (f^{-1})'(x) \\ &= x(f^{-1})'(x) + f^{-1}(x) - f(f^{-1}(x)) \cdot (f^{-1})'(x) \\ &= x(f^{-1})'(x) + f^{-1}(x) - x(f^{-1})'(x) \\ &= f^{-1}(x).\end{aligned}$$

22. (i)

$$(h^{-1})'(3) = \frac{1}{h'(h^{-1}(3))} = \frac{1}{h'(0)} = \frac{1}{\sin^2(\sin 1)}.$$

CHAPTER 13

1. If $P_n = \{t_0, \dots, t_n\}$ is the partition with $t_i = ib/n$, then

$$\begin{aligned}L(f, P_n) &= \sum_{i=1}^n (t_{i-1})^3 \cdot (t_i - t_{i-1}) \\ &= \sum_{i=1}^n (i-1)^3 \cdot \frac{b^3}{n^3} \cdot \frac{b}{n} \\ &= \frac{b^4}{n^4} \sum_{j=0}^{n-1} j^3 \\ &= \frac{b^4}{n^4} \left[\frac{(n-1)^4}{4} + \frac{(n-1)^3}{2} + \frac{(n-1)^2}{4} \right],\end{aligned}$$

and similarly

$$\begin{aligned}U(f, P_n) &= \frac{b^4}{n^4} \sum_{j=1}^n j^3 \\ &= \frac{b^4}{n^4} \left[\frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4} \right].\end{aligned}$$

Clearly $L(f, P_n)$ and $U(f, P_n)$ can be made as close to $b^4/4$ as desired by choosing n sufficiently large, so $U(f, P_n) - L(f, P_n)$ can be made as small as desired, by choosing n large enough. This shows that f is integrable. Moreover, there is only one number a with $L(f, P_n) \leq a \leq U(f, P_n)$ for all n ; since $\int_0^b x^3 dx$ has this property, the proof that $\int_0^b x^3 dx = b^4/4$ will be complete once we show that $L(f, P_n) \leq b^4/4 \leq U(f, P_n)$ for all n . This

can be done by a straightforward computation, but it actually follows from the fact that $L(f, P_n)$ and $U(f, P_n)$ can be made as close to $b^4/4$ as desired by choosing n sufficiently large. In fact, if it were true that $b^4/4 < \int_0^b x^3 dx$, then it would not be possible to make $U(f, P_n)$ as close as desired to $b^4/4$ by choosing n large enough, since each $U(f, P_n) \geq \int_0^b x^3 dx$, and similarly we cannot have $b^4/4 > \int_0^b x^3 dx$.

2. We have

$$L(f, P_n) = \frac{b^5}{n^5} \left[\frac{(n-1)^5}{5} + \frac{(n-1)^4}{2} + \frac{(n-1)^3}{3} - \frac{(n-1)}{30} \right],$$

$$U(f, P_n) = \frac{b^5}{n^5} \left[\frac{n^5}{5} + \frac{n^4}{2} + \frac{n^3}{3} - \frac{n}{30} \right].$$

Clearly $L(f, P_n)$ and $U(f, P_n)$ can be made as close to $b^5/5$ as desired by choosing n large enough. As in Problem 1, this implies that $\int_0^b x^4 dx = b^5/5$.

7. (i) $\int_0^2 f = 0$.
 (iii) $\int_0^2 f = 3$.
 (v) f is not integrable.
 (vii) $\int_0^2 f = 1$.

(For a rigorous proof that the functions in (i), (iii), and (vii) are integrable, see Problem 19. The values of the integrals, which are clear from the geometric picture, can also be deduced rigorously by using the ideas in the proof of Problem 19, together with known integrals.)

8. (i)

$$\int_{-2}^2 \left[\left(\frac{x^2}{2} + 2 \right) - x^2 \right] dx = \frac{16}{3}.$$

(iii)

$$\int_{-\sqrt{2}/2}^{\sqrt{2}/2} [(1-x^2) - x^2] dx = \frac{2\sqrt{2}}{3}.$$

(v)

$$\int_0^2 [(x^2 - 2x + 4) - x^2] dx = 4.$$

9.

$$\begin{aligned} \int_a^b \left(\int_c^d f(x)g(y) dy \right) dx &= \int_a^b \left(f(x) \int_c^d g(y) dy \right) dx \quad (\text{here } f(x) \text{ is the constant}) \\ &= \int_c^d g(y) dy \cdot \int_a^b f(x) dx \\ &\quad (\text{here } \int_c^d g(y) dy \text{ is the constant}). \end{aligned}$$

13. (a) Clearly $L(f, P) \geq 0$ for every partition P .

- (b) Apply part (a) to
- $f - g$
- , and use the fact that

$$\int_a^b (f - g) = \int_a^b f - \int_a^b g.$$

23. (a) Clearly

$$m(b - a) \leq L(f, P) \leq U(f, P) \leq M(b - a)$$

for all partitions P of $[a, b]$. Consequently,

$$m(b - a) \leq \int_a^b f(x) dx \leq M(b - a).$$

Thus

$$\mu = \frac{\int_a^b f(x) dx}{b - a}$$

satisfies $m \leq \mu \leq M$.

- (b) Let
- m
- and
- M
- be the minimum and maximum values of
- f
- on
- $[a, b]$
- . Since
- f
- is continuous, it takes on the values
- m
- and
- M
- , and consequently the number
- μ
- of part (a).

33. (a) 0.

- (b)
- $\frac{1}{2}$
- .

37. Since

$$-|f| \leq f \leq |f|,$$

we have

$$-\int_a^b |f| \leq \int_a^b f \leq \int_a^b |f|,$$

so

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

(Problem 36 implies that $\int_a^b |f|$ makes sense.)

CHAPTER 14

1. (i)
- $(\sin^3 x^3) \cdot 3x^2$
- .

$$(iii) \int_8^x \frac{1}{1 + t^2 + \sin^2 t} dt.$$

$$(v) \int_a^b \frac{1}{1 + t^2 + \sin^2 t} dt.$$

$$(vii) (F^{-1})'(x) = \frac{1}{F'(F^{-1}(x))} = F^{-1}(x).$$

2. (i) All
- $x \neq 1$
- .

- (iii) All
- $x \neq 1$
- .

- (v) All
- x
- .

- (vii) All
- $x \neq 0$
- . (
- F
- is not differentiable at 0 because
- $F(x) = 0$
- for
- $x \leq 0$
- , but there are
- $x > 0$
- arbitrarily close to 0 with
- $\frac{F(x)}{x} = \frac{1}{2}$
- .)

4. (i)

$$\begin{aligned}(f^{-1})'(0) &= \frac{1}{f'(f^{-1}(0))} = \frac{1}{1 + \sin(\sin(f^{-1}(0)))} \\ &= \frac{1}{1 + \sin(\sin 0)} = 1.\end{aligned}$$

 8. $F(x) = x \int_0^x f(t) dt$, so

$$F'(x) = xf(x) + \int_0^x f(t) dt.$$

11.

$$f(x) = \int_0^x \left(\int_0^y \left(\int_0^z \frac{1}{\sqrt{1 + \sin^2 t}} dt \right) dz \right) dy.$$

13. We can choose

$$f(x) = \frac{x^{(1/n)+1}}{\frac{1}{n} + 1}.$$

Then

$$\int_0^b \sqrt[n]{x} dx = f(b) - f(0) = \frac{b^{(1/n)+1}}{\frac{1}{n} + 1}.$$

CHAPTER 15

1. (i)

$$\frac{1}{1 + \arctan^2(\arctan x)} \cdot \frac{1}{1 + \arctan^2 x} \cdot \frac{1}{1 + x^2}.$$

(iii)

$$\frac{1}{1 + (\tan x \arctan x)^2} \cdot \left(\sec^2 x \arctan x + \frac{\tan x}{1 + x^2} \right).$$

2. (i) 0.

(iii) 0.

(v) 0.

7. (a)

$$\sin 2x = \sin(x + x) = \sin x \cos x + \cos x \sin x = 2 \sin x \cos x.$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x.$$

$$\sin 3x = \sin(2x + x) = \sin 2x \cos x + \cos 2x \sin x$$

$$= 2 \sin x \cos^2 x + (\cos^2 x - \sin^2 x) \sin x$$

$$= 3 \sin x \cos^2 x - \sin^3 x.$$

$$\cos 3x = \cos(2x + x) = \cos 2x \cos x - \sin 2x \sin x$$

$$= (\cos^2 x - \sin^2 x) \cos x - 2 \sin^2 x \cos x$$

$$= \cos^3 x - \sin^2 x \cos x - 2 \sin^2 x \cos x$$

$$= \cos^3 x - 3 \sin^2 x \cos x$$

$$= 4 \cos^3 x - 3 \cos x.$$

(b) Since $\cos \pi/4 > 0$ and

$$0 = \cos \frac{\pi}{2} = \cos 2 \cdot \frac{\pi}{4} = 2 \cos^2 \frac{\pi}{4} - 1,$$

we have $\cos \pi/4 = \sqrt{2}/2$. It follows, since $\sin \pi/4 > 0$ and $\sin^2 + \cos^2 = 1$, that $\sin \pi/4 = \sqrt{2}/2$, and consequently $\tan \pi/4 = 1$. Similarly, since $\cos \pi/6 > 0$ and

$$0 = \cos \frac{\pi}{2} = \cos 3 \cdot \frac{\pi}{6} = 4 \cos^3 \frac{\pi}{6} - 3 \cos \frac{\pi}{6},$$

we have $\cos \pi/6 = \sqrt{3}/2$. It follows, since $\sin \pi/6 > 0$, that $\sin \pi/6 = \sqrt{1 - (\sqrt{3}/2)^2} = \frac{1}{2}$.

9. (a)

$$\begin{aligned} \tan(x+y) &= \frac{\sin(x+y)}{\cos(x+y)} \\ &= \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y - \sin x \sin y} \\ &= \frac{\frac{\sin x \cos y}{\cos x \cos y} + \frac{\cos x \sin y}{\cos x \cos y}}{\frac{\cos x \cos y}{\cos x \cos y} - \frac{\sin x \sin y}{\cos x \cos y}} \\ &= \frac{\tan x + \tan y}{1 - \tan x \tan y}. \end{aligned}$$

(b) From part (a) we have

$$\begin{aligned} \tan(\arctan x + \arctan y) &= \frac{\tan(\arctan x) + \tan(\arctan y)}{1 - \tan(\arctan x) \tan(\arctan y)} \\ &= \frac{x + y}{1 - xy}, \end{aligned}$$

provided that $\arctan x$, $\arctan y$, and $\arctan x + \arctan y \neq k\pi + \pi/2$. Since $-\pi/2 < \arctan x$, $\arctan y < \pi/2$, this is always the case except when $\arctan x + \arctan y = \pm\pi/2$, which is equivalent to $xy = 1$. From this equation we can conclude that

$$\arctan x + \arctan y = \arctan \left(\frac{x+y}{1-xy} \right)$$

provided that $\arctan x + \arctan y$ lies in $(-\pi/2, \pi/2)$, which is true whenever $xy < 1$. (If $x, y > 0$ and $xy > 1$, so that $\arctan x + \arctan y > \pi/2$, then we must add π to the right side, and if $x, y < 0$ and $xy > 1$, so that $\arctan x + \arctan y < -\pi/2$, then we must subtract π .)

11. The first formula is derived by subtracting the second of the following two equations from the first:

$$\begin{aligned}\cos(m-n)x &= \cos(mx-nx) = \cos mx \cos(-nx) - \sin mx \sin(-nx) \\ &= \cos mx \cos nx + \sin mx \sin nx, \\ \cos(m+n)x &= \cos mx \cos nx - \sin mx \sin nx.\end{aligned}$$

The other formulas are derived similarly.

12. It follows from Problem 11 that if $m \neq n$, then

$$\begin{aligned}\int_{-\pi}^{\pi} \sin mx \sin nx \, dx &= \frac{1}{2} \int_{-\pi}^{\pi} [\cos(m-n)x - \cos(m+n)x] \, dx \\ &= \frac{1}{2} \left\{ \left[\frac{\sin(m-n)\pi}{m-n} - \frac{\sin(m+n)\pi}{m+n} \right] \right. \\ &\quad \left. - \left[\frac{\sin(m-n)\pi}{m-n} - \frac{\sin(m+n)\pi}{m+n} \right] \right\} \\ &= 0.\end{aligned}$$

(Note that $\sin(m-n)(-\pi) = \sin(m-n)\pi$ since $m-n$ is an integer.) But if $m = n$, then

$$\begin{aligned}\int_{-\pi}^{\pi} \sin mx \sin nx \, dx &= \frac{1}{2} \int_{-\pi}^{\pi} 1 - \cos(m+n)x \, dx \\ &= \frac{1}{2} \left\{ \left[\pi - \frac{\sin(m+n)\pi}{m+n} \right] - \left[-\pi - \frac{\sin(m+n)\pi}{m+n} \right] \right\} \\ &= \pi.\end{aligned}$$

The other formulas are proved similarly.

15. (a) We have

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x \\ &= 1 - 2\sin^2 x \\ &= 2\cos^2 x - 1.\end{aligned}$$

So

$$\begin{aligned}\sin^2 x &= \frac{1 - \cos 2x}{2}, \\ \cos^2 x &= \frac{1 + \cos 2x}{2}.\end{aligned}$$

- (b) These formulas follow from part (a), because $\cos x/2 \geq 0$ and $\sin x/2 \geq 0$ (since $0 \leq x \leq \pi/2$).

- (c)

$$\begin{aligned}\int_a^b \sin^2 x \, dx &= \int_a^b \frac{1 - \cos 2x}{2} \, dx = \frac{1}{2}(b-a) - \frac{1}{4}(\sin 2b - \sin 2a), \\ \int_a^b \cos^2 x \, dx &= \int_a^b \frac{1 + \cos 2x}{2} \, dx = \frac{1}{2}(b-a) + \frac{1}{4}(\sin 2b - \sin 2a).\end{aligned}$$

19. (a) $\arctan 1 - \arctan 0 = \pi/4$.
 (b) $\lim_{x \rightarrow \infty} \arctan x - \arctan 0 = \pi/2$.
20. $\lim_{x \rightarrow \infty} x \sin \frac{1}{x} = \lim_{x \rightarrow 0^+} \frac{1}{x} \sin x = 1$.
21. (a)

$$(\sin^\circ)'(x) = \frac{\pi}{180} \cos\left(\frac{\pi x}{180}\right) = \frac{\pi}{180} \cos^\circ(x).$$

$$(\cos^\circ)'(x) = \frac{\pi}{180} \cdot -\sin\left(\frac{\pi x}{180}\right) = \frac{-\pi}{180} \sin^\circ(x).$$
- (b) $\lim_{x \rightarrow 0} \frac{\sin^\circ x}{x} = \lim_{x \rightarrow 0} \frac{\sin(\pi x/180)}{x} = \lim_{x \rightarrow 0} \frac{\pi}{180} \cdot \frac{\sin(\pi x/180)}{\pi x/180} = \frac{\pi}{180}.$

$$\lim_{x \rightarrow \infty} x \sin^\circ \frac{1}{x} = \lim_{x \rightarrow 0^+} \frac{\sin^\circ x}{x} = \frac{\pi}{180}.$$

CHAPTER 18

1. (i) $e^{e^{e^{e^x}}} \cdot e^{e^{e^x}} \cdot e^{e^x} \cdot e^x.$
 (iii) $(\sin x)^{\sin(\sin x)} [(\log(\sin x)) \cdot \cos(\sin x) \cdot \cos x + (\cos x / \sin x) \cdot \sin(\sin x)].$
 (v) $(\sin x)^{(\sin x)^{\sin x}} \left[(\sin x)^{\sin x} \cdot \log(\sin x) \left\{ \cos x \cdot \log(\sin x) + \sin x \frac{\cos x}{\sin x} \right\} + (\sin x)^{\sin x} \cdot \frac{\cos x}{\sin x} \right].$
 (vii) $\left[\arcsin\left(\frac{x}{\sin x}\right) \right]^{\log(\sin e^x)} \left[\left(\log\left(\arcsin\left(\frac{x}{\sin x}\right)\right) \right) \cdot \frac{(\cos e^x)e^x}{\sin e^x} + \log(\sin e^x) \cdot \frac{\sin x - x \cos x}{\arcsin\left(\frac{x}{\sin x}\right) \sqrt{1 - \left(\frac{x}{\sin x}\right)^2} \cdot \sin^2 x} \right].$
 (ix) $(\log x)^{\log x} \cdot \left[\log(\log x) \cdot \frac{1}{x} + \log x \cdot \frac{1}{\log x} \cdot \frac{1}{x} \right].$
 (xi) $\cos(x^{\sin(x^{\sin x})}) \cdot x^{\sin(x^{\sin x})} \cdot \left[\cos(x^{\sin x}) \cdot x^{\sin x} \cdot \log x + \left\{ \cos x \cdot \log x + \frac{\cos x}{x} \right\} + \frac{\sin(x^{\sin x})}{x} \right].$
5. (i) 0.
 (iii) $\frac{1}{6}.$
 (v) $\frac{1}{3}.$

$$\begin{aligned}
 8. \quad (a) \quad \cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\
 &= \left[\frac{e^{2x}}{4} + \frac{1}{2} + \frac{e^{-2x}}{4} \right] - \left[\frac{e^{2x}}{4} - \frac{1}{2} + \frac{e^{-2x}}{4} \right] \\
 &= 1.
 \end{aligned}$$

(c)

$$\begin{aligned}
 \sinh x \cosh y + \cosh x \sinh y &= \left(\frac{e^x - e^{-x}}{2} \right) \left(\frac{e^y + e^{-y}}{2} \right) + \left(\frac{e^x + e^{-x}}{2} \right) \left(\frac{e^y - e^{-y}}{2} \right) \\
 &= \left[\frac{e^{x+y}}{4} - \frac{e^{-x-y}}{4} - \frac{e^{-x+y}}{4} + \frac{e^{x-y}}{4} \right] + \left[\frac{e^{x+y}}{4} - \frac{e^{-x-y}}{4} + \frac{e^{-x+y}}{4} - \frac{e^{x-y}}{4} \right] \\
 &= \frac{e^{x+y} - e^{-(x+y)}}{2} = \sinh(x+y).
 \end{aligned}$$

(e) Since

$$\sinh x = \frac{e^x - e^{-x}}{2},$$

we have

$$\sinh'(x) = \frac{e^x + e^{-x}}{2} = \cosh x.$$

(g) Since

$$\tanh x = \frac{\sinh x}{\cosh x},$$

we have

$$\begin{aligned}
 \tanh'(x) &= \frac{(\cosh x)^2 - (\sinh x)^2}{\cosh^2 x} \\
 &= \frac{1}{\cosh^2 x} \quad \text{by part (a).}
 \end{aligned}$$

9. (a) If $y = \cosh^{-1} x$, then $y \geq 0$ and

$$x = \cosh y = \sqrt{1 + \sinh^2 y} \quad \text{by Problem 7(a).}$$

So

$$\sinh(\cosh^{-1} x) = \sinh y = \sqrt{x^2 - 1} \quad \text{since } \sinh y \geq 0 \text{ for } y \geq 0.$$

(c)

$$\begin{aligned}
 (\sinh^{-1})'(x) &= \frac{1}{\sinh'(\sinh^{-1}(x))} \\
 &= \frac{1}{\cosh(\sinh^{-1}(x))} \\
 &= \frac{1}{\sqrt{1+x^2}} \quad \text{by part (b).}
 \end{aligned}$$

(e)

$$\begin{aligned}(\tanh^{-1})'(x) &= \frac{1}{\tanh'(\tanh^{-1}(x))}, \\ &= \cosh^2(\tanh^{-1}(x)).\end{aligned}$$

Now,

$$\tanh^2 y + \frac{1}{\cosh^2 y} = 1 \quad \text{by Problem 8(b),}$$

so

$$\tanh^2(\tanh^{-1}(x)) + \frac{1}{\cosh^2(\tanh^{-1}(x))} = 1,$$

or

$$\cosh^2(\tanh^{-1}(x)) = \frac{1}{1-x^2}.$$

10. (a) If $y = \sinh^{-1} x$, then

$$x = \sinh y = \frac{e^y - e^{-y}}{2}$$

so

$$\begin{aligned}e^y - e^{-y} &= 2x, \\ e^{2y} - 2xe^y - 1 &= 0, \\ e^y &= \frac{2x \pm \sqrt{4x^2 + 4}}{2}\end{aligned}$$

so

$$e^y = x + \sqrt{1+x^2} \quad \text{since } e^y > 0$$

or

$$y = \sinh^{-1} x = \log(x + \sqrt{1+x^2}).$$

Similarly,

$$\begin{aligned}\cosh^{-1} x &= \log(x + \sqrt{x^2 - 1}), \\ \tanh^{-1} x &= \frac{1}{2} \log(1+x) - \frac{1}{2} \log(1-x).\end{aligned}$$

(b)

$$\begin{aligned}\int_a^b \frac{1}{\sqrt{1+x^2}} dx &= \sinh^{-1} b - \sinh^{-1} a \quad \text{by Problem 9(c)} \\ &= \log(b + \sqrt{1+b^2}) - \log(a + \sqrt{1+a^2}).\end{aligned}$$

$$\int_a^b \frac{1}{\sqrt{x^2-1}} dx = \begin{cases} \log(b + \sqrt{b^2-1}) - \log(a + \sqrt{a^2-1}) & a, b > 1 \\ -\log(-b + \sqrt{b^2-1}) + \log(-a + \sqrt{a^2-1}) & a, b < -1 \end{cases}$$

$$\int_a^b \frac{1}{1-x^2} dx = \frac{1}{2} [\log(1+b) - \log(1-b) - \log(1+a) + \log(1-a)].$$

13. (a) $\lim_{x \rightarrow \infty} a^x = \lim_{x \rightarrow \infty} e^{x \log a}$. Since $\log a < 0$, we have $\lim_{x \rightarrow \infty} x \log a = -\infty$, so $\lim_{x \rightarrow \infty} e^{x \log a} = 0$.
- (c) $\lim_{x \rightarrow \infty} \frac{(\log x)^n}{x} = \lim_{y \rightarrow \infty} \frac{y^n}{e^y} = 0$.
- (e) $\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^{x \log x}$. Now, $\lim_{x \rightarrow 0^+} x \log x = 0$ by part (d), so $\lim_{x \rightarrow 0^+} x^x = 1$.
17. (a) $\lim_{y \rightarrow 0} \log(1+y)/y = \log'(1) = 1$.
- (b) $\lim_{x \rightarrow \infty} x \log(1+1/x) = \lim_{y \rightarrow 0^+} \log(1+y)/y = 1$.
- (c)

$$\begin{aligned} e &= \exp(1) = \exp\left(\lim_{x \rightarrow \infty} x \log(1+1/x)\right) \\ (*) &= \lim_{x \rightarrow \infty} \exp(x \log(1+1/x)) \\ &= \lim_{x \rightarrow \infty} (1+1/x)^x. \end{aligned}$$

(The starred equality depends on the continuity of $\exp$ at 1, and can be justified as follows. For every $\varepsilon > 0$ there is some $\delta > 0$ such that $|e - \exp y| < \varepsilon$ for $|y - 1| < \delta$. Moreover, there is some N such that $|x \log(1+1/x) - 1| < \delta$ for $x > N$. So $|e - \exp(x \log(1+1/x))| < \varepsilon$ for $x > N$.

(d)

$$\begin{aligned} e^a &= \left[\lim_{x \rightarrow \infty} (1+1/x)^x \right]^a = \lim_{x \rightarrow \infty} (1+1/x)^{ax} \\ &= \lim_{ax \rightarrow \infty} (1+1/x)^{ax} \\ &= \lim_{y \rightarrow \infty} (1+a/y)^y. \end{aligned}$$

19. After one year the number of dollars yielded by an initial investment of one dollar will be

$$\lim_{x \rightarrow \infty} (1+a/100x)^x = e^{a/100}.$$

20. (a) Clearly $f'(x) = 1/x$ for $x > 0$. If $x < 0$, then $f(x) = \log(-x)$, so $f'(x) = (-1) \cdot 1/(-x) = 1/x$.
- (b) We can write $\log|f|$ as $g \circ f$ where $g(x) = \log|x|$ is the function of part (a). So $(\log|f|)' = (g' \circ f) \cdot f' = 1/f \cdot f'$.
21. (c) Let $g(x) = f(x)/e^{cx}$. Then

$$g'(x) = \frac{e^{cx} f'(x) - f(x) c e^{cx}}{e^{2cx}} = 0,$$

so there is some number k such that $g(x) = k$ for all x .

22. (a) According to Problem 21, there is some k such that $A(t) = k e^{ct}$. Then $k = k e^{0 \cdot t} = A_0$. So $A(t) = A_0 e^{ct}$.
- (b) If $A(t+\tau) = A(t)/2$, then

$$A_0 e^{ct+c\tau} = \frac{A_0 e^{ct}}{2},$$

so $e^{ct}e^{c\tau} = e^{ct}/2$ or $e^{c\tau} = \frac{1}{2}$, so $\tau = -(\log 2)/c$. It is easy to check that this τ does work.

23. Newton's law states that, for a certain (positive) number c ,

$$T'(t) = c(T - M),$$

which can be written

$$(T - M)' = c(T - M).$$

So by Problem 21 there is some number k such that

$$T(t) - M = ke^{ct},$$

and $k = ke^{0t} = T(0) - M = T_0 - M$. So $T(t) = M + (T_0 - M)e^{ct}$.

CHAPTER 19

1. (i) $(\sqrt[5]{x^3} + \sqrt[6]{x})/\sqrt{x} = x^{1/10} + x^{-1/3}$.
 (ii) $\frac{1}{\sqrt{x-1} + \sqrt{x+1}} = \frac{\sqrt{x-1} - \sqrt{x+1}}{-2}$.
 (iii) $(e^x + e^{2x} + e^{3x})/e^{4x} = e^{-3x} + e^{-2x} + e^{-x}$.
 (iv) $a^x/b^x = (a/b)^x = e^{x \log(a/b)}$.
 (v) $\tan^2 x = \sec^2 x - 1$.
 (vi) $\frac{1}{a^2 + x^2} = \frac{1/a^2}{1 + \left(\frac{x}{a}\right)^2}$.
 (vii) $\frac{1}{\sqrt{a^2 - x^2}} = \frac{1/a}{\sqrt{1 - (x/a)^2}}$.
 (viii) $\frac{1}{1 + \sin x} = \frac{1 - \sin x}{1 - \sin^2 x} = \frac{1 - \sin x}{\cos^2 x} = \sec^2 x - \sec x \tan x$.
 (ix) $\frac{8x^2 + 6x + 4}{x + 1} = 8x - 2 + \frac{6}{x + 1}$.
 (x) $\frac{1}{\sqrt{2x - x^2}} = \frac{1}{\sqrt{1 - (x - 1)^2}}$.
2. (i) $-\cos e^x$. (Let $u = e^x$.)
 (iii) $(\log x)^2/2$. (Let $u = \log x$.)
 (v) e^{e^x} . (Let $u = e^{e^x}$.)
 (vii) $2e^{\sqrt{x}}$. (Let $u = \sqrt{x}$.)
 (ix) $-(\log(\cos x))^2/2$. (Let $u = \log(\cos x)$.)
3. (i) $\int x^2 e^x dx = x^2 e^x - \int 2x e^x dx = x^2 e^x - 2 \left[x e^x - \int e^x dx \right]$
 $= x^2 e^x - 2x e^x + 2e^x$.

(iii) We have

$$\begin{aligned}\int e^{ax} \sin bx \, dx &= \frac{e^{ax} \sin bx}{a} - \frac{b}{a} \int e^{ax} \cos bx \, dx \\ &= \frac{e^{ax} \sin bx}{a} - \frac{b}{a} \left[\frac{e^{ax} \cos bx}{a} - \frac{b}{a} \int e^{ax} (-\sin bx) \, dx \right],\end{aligned}$$

so

$$\int e^{ax} \sin bx \, dx = \frac{a}{a^2 + b^2} e^{ax} \sin bx - \frac{b}{a^2 + b^2} e^{ax} \cos bx.$$

(v) Using the result $\int (\log x)^2 \, dx = x(\log x)^2 - 2x(\log x) + 2x$ from the text, we have

$$\begin{aligned}\int (\log x)^3 \, dx &= [x(\log x)^2 - 2x(\log x) + 2x] \log x \\ &\quad - \int \frac{1}{x} [x(\log x)^2 - 2x(\log x) + 2x] \, dx \\ &= x(\log x)^3 - 2x(\log x)^2 + 2x \log x \\ &\quad - \int (\log x)^2 \, dx + 2[x \log x - x] - 2x \\ &= x(\log x)^3 - 2x(\log x)^2 + 2x \log x \\ &\quad - [x(\log x)^2 - 2x(\log x) + 2x] + 2[x \log x - x] - 2x \\ &= x(\log x)^3 - 3x(\log x)^2 + 6x \log x - 6x.\end{aligned}$$

(vii)

$$\begin{aligned}\int \sec^3 x \, dx &= \int (\sec^2 x)(\sec x) \, dx = \tan x \sec x - \int (\tan x)(\sec x \tan x) \, dx \\ &= \tan x \sec x - \int \sec x (\sec^2 x - 1) \, dx \\ &= \tan x \sec x - \int \sec^3 x \, dx + \int \sec x \, dx,\end{aligned}$$

so

$$\int \sec^3 x \, dx = \frac{1}{2} [\tan x \sec x + \log(\sec x + \tan x)].$$

(ix)

$$\begin{aligned}\int \sqrt{x} \log x \, dx &= \frac{2x^{3/2}}{3} \log x - \frac{2}{3} \int x^{3/2} \cdot \frac{1}{x} \, dx \\ &= \frac{2x^{3/2}}{3} \log x - \frac{2}{3} \int x^{1/2} \, dx \\ &= \frac{2x^{3/2}}{3} \log x - \frac{4}{9} x^{3/2}.\end{aligned}$$

4. (i) Let $x = \sin u$, $dx = \cos u \, du$. The integral becomes

$$\int \frac{\cos u \, du}{\sqrt{1 - \sin^2 u}} = \int 1 \, du = u = \arcsin x.$$

(iii) Let $x = \sec u$, $dx = \sec u \tan u \, du$. The integral becomes

$$\begin{aligned}\int \frac{\sec u \tan u \, du}{\sqrt{\sec^2 u - 1}} &= \int \sec u \, du = \log(\sec u + \tan u) \\ &= \log(x + \sqrt{x^2 - 1}).\end{aligned}$$

(v) Let $x = \sin u$, $dx = \cos u \, du$. The integral becomes

$$\begin{aligned}\int \frac{\cos u \, du}{\sin u \sqrt{1 - \sin^2 u}} &= \int \csc u \, du = -\log(\csc u + \cot u) \\ &= -\log\left(\frac{1}{x} + \frac{\sqrt{1 - x^2}}{x}\right).\end{aligned}$$

(vii) Let $x = \sin u$, $dx = \cos u \, du$. The integral becomes

$$\begin{aligned}\int (\sin^3 u \cos u) \cos u \, du &= \int \sin^3 u \cos^2 u \, du = \int (\sin u)(1 - \cos^2 u) \cos^2 u \, du \\ &= \int (\sin u)(\cos^2 u - \cos^4 u) \, du = -\frac{\cos^3 u}{3} + \frac{\cos^5 u}{5} \\ &= -\frac{(1 - x^2)^{3/2}}{3} + \frac{(1 - x^2)^{5/2}}{5}.\end{aligned}$$

(ix) Let $x = \tan u$, $dx = \sec^2 u \, du$. The integral becomes

$$\begin{aligned}\int \sec u \sec^2 u \, du &= \int \sec^3 u \, du \\ &= \frac{1}{2}[\tan u \sec u + \log(\sec u + \tan u)] \quad \text{by Problem 3(vii)} \\ &= \frac{1}{2}[x\sqrt{1 + x^2} + \log(x + \sqrt{1 + x^2})].\end{aligned}$$

5. (i) Let $u = \sqrt{x + 1}$, $x = u^2 - 1$, $dx = 2u \, du$. The integral becomes

$$\begin{aligned}\int \frac{2u \, du}{1 + u} &= \int \left(2 + \frac{-2}{1 + u}\right) du \\ &= 2u - 2\log(1 + u) = 2\sqrt{x + 1} - 2\log(1 + \sqrt{x + 1}).\end{aligned}$$

(iii) Let $u = x^{1/6}$, $x = u^6$, $dx = 6u^5 \, du$. The integral becomes

$$\begin{aligned}\int \frac{6u^5 \, du}{u^3 + u^2} &= 6 \int \left(u^2 - u + 1 - \frac{1}{u + 1}\right) du = 2u^3 - 3u^2 + 6u - 6\log(u + 1) \\ &= 2\sqrt[6]{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6\log(\sqrt[6]{x} + 1).\end{aligned}$$

- (v) Let $u = \tan x$, $x = \arctan u$, $dx = du/(1+u^2)$. The integral becomes

$$\begin{aligned} \int \frac{du}{(1+u^2)(2+u)} &= \frac{1}{5} \int \left(\frac{1}{2+u} - \frac{u-2}{1+u^2} \right) du \\ &= \frac{1}{5} \int \frac{du}{2+u} - \frac{1}{10} \int \frac{2u}{1+u^2} du + \frac{2}{5} \int \frac{du}{1+u^2} \\ &= \frac{1}{5} \log(2+u) - \frac{1}{10} \log(1+u^2) + \frac{2}{5} \arctan u \\ &= \frac{1}{5} \log(2+\tan x) - \frac{1}{10} \log(1+\tan^2 x) + \frac{2}{5} x. \end{aligned}$$

- (vii) Let $u = 2^x$, $x = (\log u)/(\log 2)$, $dx = du/(u \log 2)$. The integral becomes

$$\begin{aligned} \frac{1}{\log 2} \int \frac{u^2+1}{(u+1)u} du &= \frac{1}{\log 2} \int \left(1 + \frac{1-u}{u(u+1)} \right) du \\ &= \frac{1}{\log 2} \int \left(1 + \frac{1}{u} - \frac{2}{u+1} \right) du \\ &= \frac{1}{\log 2} [u + \log u - 2 \log(u+1)] \\ &= \frac{1}{\log 2} [2^x + x \log 2 - 2 \log(2^x + 1)]. \end{aligned}$$

- (ix) Let $u = \sqrt{x}$, $x = u^2$, $dx = 2u$. The integral becomes

$$\int \frac{\sqrt{1-u^2} 2u du}{1-u}.$$

Now let $u = \sin y$, $du = \cos y dy$. The integral becomes

$$\begin{aligned} \int \frac{2 \cos y \sin y \cos y}{1 - \sin y} dy &= 2 \int \frac{(1 - \sin^2 y) \sin y}{1 - \sin y} dy \\ &= 2 \int (1 + \sin y) \sin y dy \\ &= 2 \int \sin y dy + \int 1 - \cos 2y dy \\ &= -2 \cos y + y - \frac{\sin 2y}{2} = -2 \cos y + y - \sin y \cos y \\ &= -2\sqrt{1-u^2} + \arcsin u - u\sqrt{1-u^2} \\ &= -2\sqrt{1-x} + \arcsin \sqrt{x} - \sqrt{x}\sqrt{1-x}. \end{aligned}$$

The substitution $u = \sqrt{1-x}$, $x = 1-u^2$, $dx = -2u du$ leads to

$$\int \frac{-2u^2 du}{1 - \sqrt{1-u^2}}$$

and the substitution $u = \sin y$ then leads to

$$\begin{aligned}\int \frac{-2 \sin^2 y \cos y \, dy}{1 - \cos y} &= -2 \sin y - y - \sin y \cos y \\ &= -2u - \arcsin u - u\sqrt{1-u^2} \\ &= -2\sqrt{1-x} - \arcsin \sqrt{1-x} - \sqrt{1-x}\sqrt{x}.\end{aligned}$$

These answers agree, since

$$\arcsin \sqrt{x} = \frac{\pi}{2} - \arcsin \sqrt{1-x}$$

(check this by comparing their derivatives and their values for $x = 0$).

6. In these problems I will denote the original integral.

(i)

$$\begin{aligned}I &= \int \frac{2}{x-1} \, dx + \int \frac{3}{(x+1)^2} \, dx \\ &= 2 \log(x-1) - \frac{3}{x+1}.\end{aligned}$$

(iii)

$$\begin{aligned}I &= \int \frac{1}{(x-1)^2} \, dx + \int \frac{4}{(x+1)^3} \, dx \\ &= -\frac{1}{(x-1)} - \frac{2}{(x+1)^2}.\end{aligned}$$

(v)

$$\begin{aligned}I &= \frac{1}{2} \int \frac{2x}{x^2+1} \, dx + \int \frac{4}{x^2+1} \, dx \\ &= \frac{1}{2} \log(x^2+1) + 4 \arctan x.\end{aligned}$$

(vii)

$$\begin{aligned}I &= \int \frac{1}{(x+1)} \, dx + \int \frac{2x}{(x^2+x+1)} \, dx \\ &= \int \frac{1}{x+1} \, dx + \int \frac{2x+1}{x^2+x+1} \, dx - \int \frac{1}{x^2+x+1} \, dx.\end{aligned}$$

Now

$$\begin{aligned}\int \frac{1}{x^2+x+1} \, dx &= \int \frac{1}{(x+\frac{1}{2})^2 + \frac{3}{4}} \, dx \\ &= \frac{4}{3} \int \frac{1}{\left[\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right]^2 + 1} \, dx \\ &= \frac{4}{3} \cdot \frac{\sqrt{3}}{2} \arctan\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right) \\ &= \frac{2\sqrt{3}}{3} \arctan\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right),\end{aligned}$$

so

$$I = \log(x+1) + \log(x^2+x+1) - \frac{2\sqrt{3}}{3} \arctan\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right).$$

(ix)

$$\begin{aligned} I &= \int \frac{2x+1}{(x^2+x+1)^2} dx - \int \frac{1}{(x^2+x+1)^2} dx \\ &= \int \frac{2x+1}{(x^2+x+1)^2} dx - \frac{16}{9} \int \frac{1}{\left(\left[\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right]^2+1\right)^2} dx. \end{aligned}$$

Now the substitution

$$u = \frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right), \quad dx = \frac{\sqrt{3}}{2} du$$

changes the second integral to

$$-\frac{16}{9} \cdot \frac{\sqrt{3}}{2} \int \frac{du}{(u^2+1)^2}.$$

Using the reduction formula, this can be written

$$-\frac{8\sqrt{3}}{9} \left[\frac{u}{2(u^2+1)} + \frac{1}{2} \int \frac{du}{u^2+1} \right],$$

so

$$I = -\frac{1}{x^2+x+1} - \frac{\sqrt{3}(x+\frac{1}{2})}{4(x^2+x+1)} - \frac{4\sqrt{3}}{9} \arctan\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right).$$

14. The equation $\int e^x \sin x \, dx = e^x \sin x - e^x \cos x - \int e^x \sin x \, dx$ means that any function F with $F'(x) = e^x \sin x$ can be written $F(x) = e^x \sin x - e^x \cos x - G(x)$ where G is another function with $G'(x) = e^x \sin x$. Of course, $G = F + c$ for some number c , but it is not necessarily true that $F = G$.

16. (a)

$$\begin{aligned} \int \arcsin x \, dx &= \int 1 \cdot \arcsin x \, dx = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx \\ &= x \arcsin x + \sqrt{1-x^2}. \end{aligned}$$

17. (a)

$$\begin{aligned}
\int \sin^4 x \, dx &= -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \int \sin^2 x \, dx \\
&= -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \left[-\frac{\sin x \cos x}{2} + \frac{1}{2} \int 1 \, dx \right] \\
&= -\frac{\sin^3 x \cos x}{4} - \frac{3 \sin x \cos x}{8} + \frac{3}{8} x. \\
\int \sin^4 x \, dx &= \int \left(\frac{1 - \cos 2x}{2} \right)^2 dx = \int \left(\frac{1}{4} - \frac{\cos 2x}{2} + \frac{\cos^2 2x}{4} \right) dx \\
&= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{1}{4} \int \frac{1 + \cos 4x}{2} dx \\
&= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{1}{4} \left[\frac{x}{2} + \frac{\sin 4x}{8} \right] \\
&= \frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32}.
\end{aligned}$$

(b) It follows that these two answers are the same, since they have the same value for $x = 0$.

21. (a)

$$\begin{aligned}
\sin^n x \, dx &= \int (\sin x)(\sin^{n-1} x) \, dx \\
&= -\cos x \sin^{n-1} x + (n-1) \int \cos x (\sin^{n-2} x) \cos x \, dx \\
&= -\cos x \sin^{n-1} x + (n-1) \int (\sin^{n-2} x - \sin^n x) \, dx,
\end{aligned}$$

so

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx.$$

(b)

$$\begin{aligned}
\int \cos^n x \, dx &= \int (\cos x)(\cos^{n-1} x) \, dx \\
&= \sin x \cos^{n-1} x + (n-1) \int \sin x (\cos^{n-2} x) \sin x \, dx \\
&= \sin x \cos^{n-1} x + (n-1) \int (\cos^{n-2} x - \cos^n x) \, dx,
\end{aligned}$$

so

$$\int \cos^n x \, dx = \frac{1}{n} \sin x \cos^{n-1} x + \frac{n-1}{n} \int \cos^{n-2} x \, dx.$$

(c)

$$\begin{aligned}\int \frac{dx}{(x^2+1)^n} &= \int \frac{dx}{(x^2+1)^{n-1}} - \int \frac{x^2 dx}{(x^2+1)^n} \\ &= \int \frac{dx}{(x^2+1)^{n-1}} - \int x \cdot \frac{x}{(x^2+1)^n} dx \\ &= \int \frac{dx}{(x^2+1)^{n-1}} - \left[\frac{x}{2(1-n)(x^2+1)^{n-1}} \right. \\ &\quad \left. - \int \frac{dx}{2(1-n)(x^2+1)^{n-1}} \right]\end{aligned}$$

so

$$\int \frac{dx}{(x^2+1)^n} = \frac{1}{2(n-1)} \frac{x}{(x^2+1)^{n-1}} - \frac{2n-3}{2(n-1)} \int \frac{1}{(x^2+1)^{n-1}} dx.$$

We can also use the substitution $x = \tan u$, $dx = \sec^2 u du$, which changes the integral to

$$\begin{aligned}\int \frac{\sec^2 u du}{\sec^{2n} u} &= \int \cos^{2n-2} u du \\ &= \frac{1}{2n-2} \cos^{2n-3} u \sin u + \frac{2n-3}{2n-2} \int \cos^{2n-4} u du \\ &= \frac{1}{2n-2} \cdot \frac{1}{(\sqrt{x^2+1})^{2n-3}} \cdot \frac{x}{\sqrt{x^2+1}} + \frac{2n-3}{2n-2} \int \frac{dx}{(x^2+1)^{n-1}} \\ &= \frac{1}{2(n-1)} \frac{x}{(x^2+1)^{n-1}} + \frac{2n-3}{2n-2} \int \frac{dx}{(x^2+1)^{n-1}}.\end{aligned}$$

CHAPTER 20

1. (i) $P_{3,0}(x) = e + ex + ex^2 + (5e/3!)x^3$.
- (iii) $P_{2n,\pi/2}(x) = 1 - \frac{(x-\pi/2)^2}{2!} + \frac{(x-\pi/2)^4}{4!} - \cdots + \frac{(-1)^n(x-\pi/2)^{2n}}{(2n)!}$.
- (v) $P_{n,1}(x) = e + e(x-1) + \frac{e(x-1)^2}{2!} + \cdots + \frac{e(x-1)^n}{n!}$.
- (vii) $P_{4,0}(x) = x + x^3$.
- (ix) $P_{2n+1,0}(x) = 1 - x^2 + x^4 - \cdots + (-1)^n x^{2n}$.
2. If f is a polynomial function of degree n , then $f^{(n+1)} = 0$. It follows from Taylor's Theorem that $R_{n,a}(x) = 0$, so $f(x) = P_{n,a}(x)$.
 - (i) $-12 + 2(x-3) + (x-3)^2$.
 - (iii) $243 + 405(x-3) + 270(x-3)^2 + 90(x-3)^3 + 15(x-3)^4 + (x-3)^5$.
3. (i) $\sum_{i=0}^9 \frac{(-1)^i}{(2i+1)!} \left(\text{since } \frac{1}{(2n+2)!} < 10^{-17} \text{ for } 2n+2 \geq 19, \text{ or } n \geq 9 \right)$.
- (iii) $\sum_{i=0}^8 \frac{(-1)^i}{2^i(2i+1)!} \left(\text{since } \frac{1}{2^{2n+2}(2n+2)!} < 10^{-20} \text{ for } 2n+2 \geq 18, \right.$
 $\left. \text{or } n \geq 8 \right)$.

- (v) $\sum_{i=0}^{13} \frac{2^i}{i!} \left(\text{since } \frac{3^2 \cdot 2^{n+1}}{(n+1)!} < 10^{-5} \text{ for } n+1 \geq 14, \text{ or } n \geq 13 \right).$
7. (i) $c_i = a_i + b_i.$
 (iii) $c_i = (i+1)a_i.$
 (v) $c_0 = \int_0^a f(t) dt; c_i = a_{i-1}/i \text{ for } i > 0.$

CHAPTER 22

1. (i) $1 - n/(n+1) = 1/(n+1) < \varepsilon \text{ for } n+1 > 1/\varepsilon.$
 (iii) $\lim_{n \rightarrow \infty} \sqrt[8]{n^2+1} - \sqrt[4]{n+1} = \lim_{n \rightarrow \infty} (\sqrt[8]{n^2+1} - \sqrt[8]{n^2}) + \lim_{n \rightarrow \infty} (\sqrt[4]{n} - \sqrt[4]{n+1}) = 0 + 0 = 0.$ (Each of these two limits can be proved in the same way that $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$ was proved in the text.)
 (v) Clearly $\lim_{n \rightarrow \infty} (\log a)/n = 0.$ So $\lim_{n \rightarrow \infty} \sqrt[n]{a} = \lim_{n \rightarrow \infty} e^{(\log a)/n} = e^0$ (by Theorem 1) = 1.
 (vii) $\sqrt[n]{n^2} \leq \sqrt[n]{n^2+n} \leq \sqrt[n]{2n^2}$, so $(\sqrt[n]{n})^2 \leq \sqrt[n]{n^2+n} \leq \sqrt[n]{2}(\sqrt[n]{n})^2$, and $\lim_{n \rightarrow \infty} (\sqrt[n]{n})^2 = \lim_{n \rightarrow \infty} \sqrt[n]{2}(\sqrt[n]{n})^2 = 1$ by parts (v) and (vi).
 (ix) Clearly $\alpha(n) \leq \log_2 n$, and $\lim_{n \rightarrow \infty} (\log_2 n)/n = 0.$
5. (a) If $0 < a < 2$, then $a^2 < 2a < 4$, so $a < \sqrt{2a} < 2.$
 (b) Part (a) shows that

$$\sqrt{2} < \sqrt{2\sqrt{2}} < \sqrt{2\sqrt{2\sqrt{2}}} < \dots < 2,$$

so the sequence converges by Theorem 2.

- (c) If this sequence is denoted by $\{a_n\}$, then the sequence $\{\sqrt{2a_n}\}$ is the same as $\{a_{n+1}\}$. So the hint shows that $l = \sqrt{2l}$, or $l = 2.$
8. If x is rational, then $n!\pi x$ is a multiple of π for sufficiently large n , so $(\cos n!\pi x)^{2k} = 1$ for all such n , so $\lim_{n \rightarrow \infty} (\lim_{k \rightarrow \infty} (\cos n!\pi x)^{2k}) = 1.$ If x is irrational, then $n!\pi x$ is not a multiple of π for any n , so $|\cos n!\pi x| < 1$, so $\lim_{k \rightarrow \infty} (\cos n!\pi x)^{2k} = 0$, so $f(x) = 0.$
9. (i) $\int_0^1 e^x dx = e - 1.$ (Use partitions of $[0, 1]$ into n equal parts.)
 (iii) $\int_0^1 \frac{1}{1+x} dx = \log 2.$
 (v) $\int_0^1 \frac{1}{(1+x)^2} dx = \frac{1}{2}.$

CHAPTER 23

1. (i) (Absolutely) convergent, since $|(\sin n\theta)/n^2| \leq 1/n^2.$
 (iii) Divergent, since the first $2n$ terms have sum $\frac{1}{2} + \dots + 1/n.$ (Leibniz's Theorem does not apply since the terms are not decreasing in absolute value.)

(v) Divergent, since

$$\frac{1}{\sqrt[3]{n^2-1}} \geq \frac{1}{n^{2/3}}.$$

(vii) Convergent, since

$$\lim_{n \rightarrow \infty} \frac{(n+1)^2/(n+1)!}{n^2/n!} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^2 \cdot \frac{1}{n+1} = 0.$$

(ix) Divergent, since $1/(\log n) > 1/n$.

(xi) Convergent, since $1/(\log n)^n < \frac{1}{2^n}$ for $n > 9$.

(xiii) Divergent, since

$$\frac{n^2}{n^3+1} > \frac{1}{2n}$$

for large enough n .

(xv) Divergent, since

$$\int_2^N \frac{1}{x \log x} dx = \log(\log N) - \log(\log 2) \rightarrow \infty \text{ as } N \rightarrow \infty.$$

(Notice that $f(x) = 1/(x \log x)$ is decreasing on $[2, \infty)$, since

$$f'(x) = \frac{-(1 + \log x)}{(x \log x)^2} < 0 \quad \text{for } x > 1.$$

(xvii) Convergent, since $1/n^2(\log n) < 1/n^2$ for $n > 2$.

(xix) Convergent, since

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{2^{n+1}(n+1)!/(n+1)^{n+1}}{2^n n!/n^n} &= \lim_{n \rightarrow \infty} \frac{2(n+1)n^n}{(n+1)^{n+1}} \\ &= \lim_{n \rightarrow \infty} \frac{2}{\left(1 + \frac{1}{n}\right)^n} = \frac{2}{e}, \end{aligned}$$

by Problem 18-17.

7. (a) For each N we clearly have

$$0 \leq \sum_{n=1}^N a_n 10^{-n} < 9 \sum_{n=1}^{\infty} 10^{-n} = 1,$$

so $\sum_{n=1}^{\infty} a_n 10^{-n}$ converges by the boundedness criterion, and lies between 0 and 1. (Actually, this number is denoted by $0.a_1a_2a_3a_4 \dots$ only when the sequence $\{a_n\}$ is not eventually 0.)

20. The area of the shaded region is $\frac{1}{2}$. The integral is

$$\begin{aligned}
 & \frac{1}{2}([1 - \frac{1}{2}] + [\frac{1}{4} - \frac{1}{8}] + [\frac{1}{16} - \frac{1}{32}] + \cdots) - \frac{1}{2}([\frac{1}{2} - \frac{1}{4}] + [\frac{1}{8} - \frac{1}{16}] + \cdots) \\
 &= \frac{1}{2}(\frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \frac{1}{128} + \cdots) - \frac{1}{2}(\frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \cdots) \\
 &= \frac{1}{4}(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots) - \frac{1}{8}(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots) \\
 &= \frac{1}{8}\left(1 + \frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \cdots\right) \\
 &= \frac{1}{8} \cdot \frac{1}{1 - \frac{1}{4}} \\
 &= \frac{1}{6}.
 \end{aligned}$$

CHAPTER 24

1. (i)

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0, & x = 0 \\ 1, & 0 < x \leq 1. \end{cases}$$

$\{f_n\}$ does not converge uniformly to f .

- (iii) $f(x) = \lim_{n \rightarrow \infty} f_n(x) = 0$ (since $\lim_{n \rightarrow \infty} x^n = \infty$ for $x > 1$). The sequence $\{f_n\}$ does not converge uniformly to f ; in fact, for any n we have $f_n(x)$ large for sufficiently large x .

- (v) $f(x) = \lim_{n \rightarrow \infty} f_n(x) = 0$, and $\{f_n\}$ converges uniformly to f , since $|f_n(x)| \leq 1/n$ for all x .

3. (i) $-\frac{1}{a} - \frac{x}{a^2} - \frac{x^2}{a^3} - \cdots$.

(iii) $\sum_{k=0}^{\infty} (-1)^k \binom{-\frac{1}{2}}{k} x^k$.

(v) $\sum_{k=0}^{\infty} \frac{(-1)^k \binom{-\frac{1}{2}}{k}}{2k+1} x^{2k+1}$.

4. (i) e^{-x} .

- (iii) If

$$f(x) = \frac{x^2}{2} - \frac{x^3}{3 \cdot 2} + \frac{x^4}{4 \cdot 3} - \cdots, \quad |x| \leq 1$$

then

$$f'(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

$$= \log(1+x) \quad |x| < 1,$$

so for $|x| < 1$ we have $f(x) = (1+x)\log(1+x) - (1+x) + c$ for some number c . Since $f(0) = 0$, we have $c = 1$, so $f(x) = (1+x) \cdot \log(1+x) - x$ for $|x| < 1$.

6. Since

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

we have

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!}$$

(notice that the right side is 1 for $x = 0$). So

$$f^{(k)}(0) = \begin{cases} \frac{(-1)^n}{2n+1}, & k = 2n \\ 0, & k \text{ odd.} \end{cases}$$

CHAPTER 25

1. (i) $|3 + 4i| = 5$; $\theta = \arctan \frac{4}{3}$.
- (iii) $|(1+i)^5| = (|1+i|)^5 = (\sqrt{2})^5$; since $\pi/4 = \arctan 1/1$ is an argument for $1+i$, an argument for $(1+i)^5$ is $5\pi/4$.
- (v) $|(|3+4i|)| = |5| = 5$; $\theta = 0$.
2. (i)

$$\begin{aligned} x &= \frac{-i \pm \sqrt{-1-4}}{2} \\ &= \frac{-i \pm \sqrt{5}i}{2} \\ &= \frac{(-1 + \sqrt{5})i}{2} \quad \text{or} \quad \frac{(-1 - \sqrt{5})i}{2}. \end{aligned}$$

- (iii) $x^2 + 2ix - 1 = (x+i)^2$, so the only solution is $x = -i$.
- (v) $x^3 - x^2 - x - 2 = (x-2)(x^2 + x + 1)$. The solutions are

$$2, \quad -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

3. (i) All $z = iy$ with y real.
- (iii) All z on the perpendicular bisector of the line segment between a and b .
- (v) For $z = x + iy$, we need $\sqrt{x^2 + y^2} < 1 - x$. This requires that $1 - x > 0$, and then our inequality is equivalent to $x^2 + y^2 < (1 - x)^2$, or $x < (1 - y^2)/2$ (and conversely this inequality implies that $x < \frac{1}{2}$, so that $1 - x > 0$ holds). The set of points $x + iy$ with $x = (1 - y^2)/2$ is the parabola pointing along the second axis, with the point $\frac{1}{2} + 0i$ closest to the origin, and which passes through the points $0 + i$ and $0 - i$; the desired set of complex numbers is the set of points inside this parabola.
4. $|x + iy|^2 = x^2 + y^2 = x^2 + (-y)^2 = |x - iy|^2$.
 $(z + \bar{z})/2 = [(x + iy) + (x - iy)]/2 = x$.
 $(z - \bar{z})/2 = [(x + iy) - (x - iy)]/2i = y$.
5. $|z + w|^2 + |z - w|^2 = (z + w)(\bar{z} + \bar{w}) + (z - w)(\bar{z} - \bar{w}) = 2z\bar{z} + 2w\bar{w} = 2(|z|^2 + |w|^2)$. Geometrically, this says that the sum of the squares of the diagonals of a parallelogram equal the sum of the squares of the sides.

CHAPTER 27

1. (i) Converges absolutely, since $|(1+i)^n/n!| = (\sqrt{2})^n/n!$, and $\sum_{n=1}^{\infty} (\sqrt{2})^n/n!$ converges.

- (iii) Converges, but not absolutely, since the real terms form the series

$$-\frac{1}{2} + \frac{1}{4} - \frac{1}{6} + \frac{1}{8} - \cdots$$

and the imaginary terms form the series

$$i\left(\frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right).$$

- (v) Diverges, since the real terms form the series

$$\frac{\log 3}{3} + 2\frac{\log 4}{4} + \frac{\log 5}{5} + \frac{\log 7}{7} + 2\frac{\log 8}{8} + \frac{\log 9}{9} + \cdots.$$

2. (i) The limit

$$\lim_{n \rightarrow \infty} \frac{|z|^{n+1}/(n+1)^2}{|z|^n/n^2} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right)^2 |z| = |z|$$

is < 1 for $|z| < 1$, but > 1 for $|z| > 1$.

- (iii) The limit

$$\lim_{n \rightarrow \infty} \frac{|z|^{n+1}}{|z|^n} = |z|$$

is < 1 for $|z| < 1$ but > 1 for $|z| > 1$.

- (v) The limit

$$\lim_{n \rightarrow \infty} \frac{2^{n+1}|z|^{(n+1)!}}{2^n|z|^{n!}} = \lim_{n \rightarrow \infty} 2|z|^{(n+1)!-n!}$$

is 0 for $|z| < 1$, but ∞ for $|z| > 1$.

3. (i) The limits

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{|z|^{2n}}{3^n}} = \frac{|z|}{\sqrt{3}} \quad \text{and} \quad \lim_{n \rightarrow \infty} \sqrt[n]{\frac{|z|^{2n+1}}{2^{n+1}}} = \frac{|z|}{\sqrt{2}}$$

are < 1 for $|z| < \sqrt{2}$, so the series converges absolutely for $|z| < \sqrt{2}$. But the series does not converge absolutely for $|z| > \sqrt{2}$, so the radius of convergence is $\sqrt{2}$.

- (iii) Since

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{n|z|^n}{2^n}} = \lim_{n \rightarrow \infty} \frac{|z|}{2} \sqrt[n]{n} = \frac{|z|}{2} \quad \text{by Problem 22-1(vi),}$$

the radius of convergence is 2.

- (v) The limit

$$\lim_{n \rightarrow \infty} \sqrt[n]{2^n z^{n!}} = 2 \lim_{n \rightarrow \infty} z^{(n-1)!}$$

is 0 for $|z| < 1$, but ∞ for $|z| > 1$, so the radius of convergence is 1.

GLOSSARY OF SYMBOLS

